# DP-ADAGE: Differentially Private Active Defenses Against GNN Extraction

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**Abstract**

Graph Neural Networks (GNNs) served through prediction APIs are vulnerable to model extraction attacks, in which repeated queries let an adversary train a surrogate that copies the victim. Existing defenses, such as ADAGE, are heuristic and cannot bound leakage over long sessions. This paper presents DP-ADAGE, a defense with a provable inference-time Rényi Differential Privacy (RDP) guarantee. DP-ADAGE counts the graph communities each user has queried and uses this diversity score to set a per-query privacy budget through a dead-zone sigmoid allocation. Once the total budget is exhausted, the API returns outputs that do not depend on the query. The full transcript then satisfies a Differential Privacy guarantee that depends only on the total budget, the Rényi order, and the failure probability. On five graph benchmarks with 200 queries per attacker and a total budget of 200, DP-ADAGE lowers the average surrogate accuracy from $\mathbf{79.9 \pm 0.5}\%$ to $\mathbf{34.5 \pm 2.6}\%$, and attacker fidelity falls to $\mathbf{35.5}\%$. Benign accuracy stays within 0.1 points of the undefended baseline, at 1.3 to 2.1 times the query latency. DP-ADAGE gives the lowest attacker accuracy on every dataset, and paired $\boldsymbol{t}$-tests show significant differences from all four baselines ($\boldsymbol{p < 10^{-4}}$). This large budget gives only a weak formal guarantee, but tighter budgets of 1 and 10 lower the average surrogate accuracy to $\mathbf{22.0 \pm 2.4}\%$ and $\mathbf{20.5 \pm 1.0}\%$.

**Keywords:** Graph neural networks, Model extraction attacks, Rényi Differential Privacy, Active defense, Query diversity; Randomized Response

# 1 Introduction

Graph Neural Networks (GNNs) [1] have become a widely used approach for learning from relational data, driving critical applications in social network analysis, drug discovery, and fraud detection. Because training high-performing GNNs requires massive, often proprietary graph datasets and significant computational resources, the resulting models hold substantial intellectual property (IP) value. Consequently, GNNs deployed as cloud-based prediction APIs are prime targets for *model extraction attacks*. Such attacks were first demonstrated against standard machine-learning APIs by Tramèr et al. [2]. In a typical attack, an adversary issues a series of black-box queries to the victim API and uses the responses to train a surrogate model, effectively stealing the victim's functionality without ever accessing its underlying parameters or training data. Follow-up work has shown that such attacks can reach very high fidelity, closely matching the decision boundary of the victim model [3].

ADAGE was recently proposed by Xu et al. [4] as the first active defense mechanism tailored specifically against GNN model extraction. ADAGE operates by monitoring the structural diversity of a user's queries and actively perturbing the API responses when the queries span an unusually large number of graph communities. While empirically effective against naive extraction attempts, ADAGE suffers from an important theoretical limitation: it relies entirely on heuristic noise addition and label-flipping without any formal privacy guarantees. Because it lacks a rigorous mathematical foundation, ADAGE is vulnerable to adaptive attackers who can optimize their queries to evade its heuristic thresholds, and it provides no composition bounds to quantify or limit the cumulative information leakage over an extended session of queries.

To overcome these limitations, this paper introduces DP-ADAGE, a novel defense framework that bridges the gap between active GNN defenses and formal privacy theory. DP-ADAGE integrates the structural intuition of community-aware active defenses with the rigorous mathematical foundations of Rényi Differential Privacy (RDP) [5]. By employing a community-aware budget allocation mechanism, DP-ADAGE dynamically maps a user's query diversity score to a formal RDP privacy budget, which directly sets the inference-time noise scale. Furthermore, by explicitly clipping output logits and embeddings to strictly bound their $\ell_2$-sensitivity, DP-ADAGE smoothly transitions between two regimes: high-utility, low-noise responses for benign users querying localized neighborhoods, and calibrated Gaussian noise that drives the signal-to-noise ratio down for attackers probing the broader graph structure.

In summary, this paper makes the following main contributions:

- **Formal Privacy Guarantees:** This paper presents an active GNN defense that provides an $(\varepsilon, \delta)$-Differential Privacy guarantee at inference time, under the specified adjacency relation and mechanism assumptions, as a defense against model extraction attacks. The guarantee is underpinned by RDP composition over the

entire query transcript and a strict Noise Floor mechanism. This mechanism enforces maximum perturbation once the cumulative privacy budget is exhausted, ensuring that post-exhaustion responses are independent of the query under the stated mechanism.

- **Adaptive Threat Modeling:** This paper designs a community-aware budget allocation function that dynamically distinguishes benign, localized user queries from the diverse, graph-spanning query patterns characteristic of extraction attacks.
- **Rigorous Utility Analysis:** This paper establishes a formal mathematical framework for the privacy-utility trade-off. By analytically bounding the adaptive noise scale $\sigma(\tau)$ injected into the API responses, this analysis guarantees high downstream classification accuracy for benign users while strictly bounding the information available to surrogate models.
- **Comprehensive Evaluation:** This paper provides extensive empirical validation demonstrating that DP-ADAGE achieves a favorable empirical privacy–utility trade-off in the evaluated settings: with a total privacy budget of $\varepsilon_{\text{total}} = 200.0$, it drives attacker surrogate accuracy down to an average of 32.2% across the five benchmark datasets, ranging from 25.2% on Cora to 43.9% on Pubmed, while simultaneously preserving an average of 80.5% downstream accuracy for benign users, within 0.1 points of the undefended baseline on every dataset.

# 2 Background and Related Work

## 2.1 Graph Neural Networks

A GNN learns a mapping $f_\theta : \mathcal{G} \times \mathcal{X} \to \mathcal{Y}$ over an attributed graph $G = (\mathcal{V}, \mathcal{E}, \mathbf{X})$, where $\mathcal{V}$ is the node set, $\mathcal{E}$ the edge set, and $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times d}$ the node feature matrix. A standard $L$-layer GNN updates node representations via

$$\mathbf{h}_v^{(\ell)} = \phi\Big(\mathbf{h}_v^{(\ell-1)},\ \mathsf{AGG}\Big(\Big\{\mathbf{h}_u^{(\ell-1)} : u \in \mathcal{N}(v)\Big\}\Big)\Big), \tag{1}$$

where $\phi$ is a learnable transformation, $\mathsf{AGG}$ is a permutation-invariant aggregation function, and $\mathcal{N}(v)$ is the neighborhood of $v$ [6]. The final representation $\mathbf{h}_v^{(L)}$ is used for node classification [1], link prediction [7], or graph-level tasks [8]. In the API deployment model considered in this paper, the victim model $f_\theta$ is served through an inference endpoint that accepts a node index and returns either class-probability logits or the penultimate-layer embedding.

## 2.2 Model Extraction Attacks on GNNs

Model extraction (also called *model stealing*) was introduced by [2] for ML classifiers and has since been specialized to GNNs. The attacker's goal is to train a surrogate model $\hat{f}_{\hat{\theta}}$ that approximates the victim model $f_\theta$. To do so, the attacker issues repeated API queries and fits the surrogate to the responses, without accessing $\theta$ or the training data.

Existing GNN extraction attacks differ mainly in their query strategy. The simplest approach is random querying [2], where nodes are sampled uniformly at random

from an unlabeled pool. A more sample-efficient alternative is active-learning-based querying [3], in which the attacker uses uncertainty sampling or coreset selection to maximize the information gain of each query. The strategy most relevant to this paper is graph-topology-aware querying [9], where the attacker exploits structural properties of the graph, such as selecting nodes from diverse neighborhoods, to reconstruct the victim's decision boundaries across the entire graph. This strategy is particularly effective because a GNN's output for a node depends on its $L$-hop subgraph, so an attacker who probes structurally diverse nodes spanning many communities can cover a much larger part of the victim's input distribution with the same query budget [10], yielding a surrogate of high fidelity. This structural diversity is precisely what DP-ADAGE's *query diversity score* $\tau$ is designed to detect and penalize.

The ADAGE framework [4] is the first *active defense* tailored to GNN extraction. It detects community-spanning queries and applies heuristic label-flipping or noise scaling proportional to the community-coverage fraction $\tau$. In the experiments reported in this paper, ADAGE reduces surrogate accuracy from 0.85 to 0.70; however, this residual accuracy of 0.70 is still substantially above the random-guess baseline of $\approx 0.14$ for Cora's 7-class task, and, critically, ADAGE provides no formal guarantee on the information leakage. DP-ADAGE inherits ADAGE's structural intuition and provides the missing formal foundation.

## 2.3 Differential Privacy

A randomized mechanism $\mathcal{M} : \mathcal{D} \to \mathcal{R}$ satisfies $(\varepsilon, \delta)$-DP [11] if for all adjacent datasets $D, D'$ (differing in one record) and all measurable output sets $S \subseteq \mathcal{R}$, Equation 2 holds:

$$\Pr[\mathcal{M}(D) \in S] \;\leq\; e^{\varepsilon} \Pr[\mathcal{M}(D') \in S] + \delta. \tag{2}$$

The parameter $\varepsilon$ bounds the log-likelihood ratio between outputs on adjacent inputs, and $\delta$ is a failure probability.

A closely related notion is Rényi Differential Privacy (RDP) [5], which gives tighter composition bounds than standard DP and has become the standard tool for tracking privacy budgets in deep learning [12]. A mechanism $\mathcal{M}$ satisfies $(\alpha, \varepsilon)$-RDP if for all adjacent $D, D'$, Equation 3 holds:

$$D_{\alpha}(\mathcal{M}(D) \,\|\, \mathcal{M}(D')) \;\leq\; \varepsilon, \tag{3}$$

where $D_{\alpha}(P\|Q) = \frac{1}{\alpha-1} \log \mathbb{E}_{x\sim Q}\left[(P(x)/Q(x))^{\alpha}\right]$ is the Rényi divergence of order $\alpha > 1$.

The mechanisms in this paper rely on the Gaussian version of this bound. Let $f$ be an inference-time output function that returns logits or embeddings. Output clipping bounds its $\ell_2$-sensitivity $\Delta_2$, independent of any property of the model parameters. For an output function with bounded $\ell_2$-sensitivity, the Gaussian mechanism $\mathcal{M}(x) = \mathrm{clip}(f(x), C) + \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$ satisfies $\left(\alpha,\ \frac{\alpha\Delta_2^2}{2\sigma^2}\right)$-RDP for all $\alpha > 1$ [5]. Conversely, a target $(\alpha, \varepsilon)$-RDP guarantee requires setting the noise scale to $\sigma = \sqrt{\alpha\Delta_2^2/(2\varepsilon)}$.

RDP guarantees also compose cleanly across multiple queries. If $\mathcal{M}_1$ and $\mathcal{M}_2$ satisfy $(\alpha, \varepsilon_1)$-RDP and $(\alpha, \varepsilon_2)$-RDP respectively, their adaptive composition satisfies $(\alpha, \varepsilon_1 + \varepsilon_2)$-RDP [5]. This additive composition rule is what makes exact budget tracking across queries possible, and it is the property the RDPAccountant in Section 3.3 relies on.

Finally, an RDP guarantee can always be converted back into a standard $(\varepsilon, \delta)$-DP guarantee for reporting. An $(\alpha, \varepsilon_{\text{RDP}})$-RDP mechanism satisfies $\left(\varepsilon_{\text{RDP}} + \frac{\log(1/\delta)}{\alpha - 1}, \delta\right)$-DP for any $\delta \in (0, 1)$ [13].

### 2.4 Related Work on Privacy-Preserving GNN Inference

Several works apply DP-SGD [12] during GNN *training* [14, 15] to protect individual training nodes. DP-SGD bounds gradient sensitivity by clipping the gradients during backpropagation. Training-time DP, however, does not protect the model's functional outputs from being extracted through black-box queries. It gives membership-inference protection for the training set. DP-ADAGE instead operates at *inference time*. Rather than clipping gradients, it clips the API outputs, logits or embeddings, directly. This bounds the output sensitivity and limits the information that the response transcript reveals, under the assumptions stated in Section 4. Resistance to model extraction is valuated empirically. Training-time DP and inference-time DP address different threat models, so this paper does not compare them experimentally.

A different line of work adds fixed Gaussian noise to all API responses [16]. This approach applies the same noise level to every user, so benign and malicious users receive the same degradation. In the experiments of this paper, static noise ($\sigma = 0.05$) preserved benign accuracy but gave almost no protection, because the surrogate accuracy stayed close to the undefended value. DP-ADAGE keeps benign accuracy within 0.1 points of the undefended model and lowers the surrogate accuracy substantially.

Other systems monitor the query stream and restrict users who exceed a threshold [17]. These approaches lack formal privacy guarantees, since a patient attacker who spreads queries over time can stay below the threshold. They also do not distinguish between benign high-volume users and attackers.

A separate line of defense uses model watermarking [18], which embeds verifiable signatures into the victim model to detect stolen surrogates after the fact. Unlike DP-ADAGE, watermarking does not *prevent* extraction. It only enables attribution after a theft has happened.

Finally, cryptographic methods such as secure inference with homomorphic encryption [19] or multi-party computation [20] provide cryptographic security guarantees for the computation. They also incur an overhead of orders of magnitude in latency, which makes them unsuitable for production GNN APIs.

## 3 DP-ADAGE System Design

Figure 1 gives an overview of the DP-ADAGE pipeline. Every query that reaches the victim API passes through four sequential components: the *CommunityTracker*, the

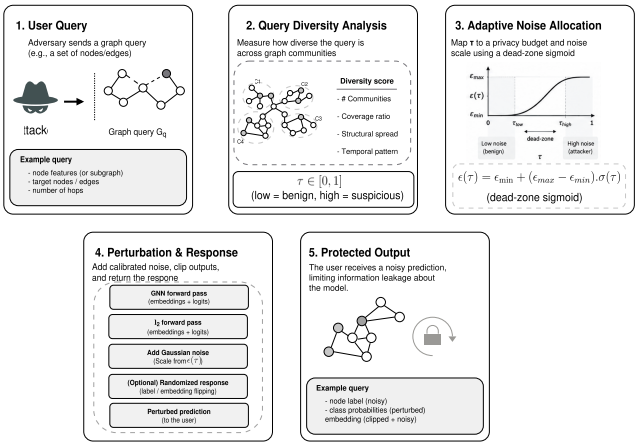



**Fig. 1 Overview of the proposed DP-ADAGE framework.** Each query is first analyzed to measure its community diversity, which sets an adaptive privacy budget. The RDP accountant tracks the cumulative budget, and calibrated noise protects the GNN output. The framework returns a privacy-preserving response and keeps utility high for benign users.

*BudgetAllocator*, the *RDPAccountant*, and the *Perturbation Mechanism*. Each component is described below. The full query-response protocol that connects them is given at the end of this section.

## 3.1 CommunityTracker

The CommunityTracker splits the training graph $G$ into $K$ communities $\{\mathcal{C}_1, \ldots, \mathcal{C}_K\}$. It uses the Louvain algorithm [21] in a one-time offline preprocessing step. The default configuration uses $K = 10$. For each community $\mathcal{C}_k$, the tracker computes a centroid embedding, as defined in Equation 4:

$$\boldsymbol{\omega}_k = \frac{1}{|\mathcal{C}_k|} \sum_{v \in \mathcal{C}_k} \mathbf{e}_v, \quad k = 1, \ldots, K, \tag{4}$$

where $\mathbf{e}_v = f_\theta^{(L-1)}(v)$ is the GNN embedding of node $v$ at the penultimate layer.

***Per-Query Community Assignment.***

When user $u$ submits a query that yields embedding $\mathbf{e}$, the tracker finds the nearest community with Equation 5:

$$k^* = \arg\min_{k \in [K]} \|\mathbf{e} - \boldsymbol{\omega}_k\|_2. \tag{5}$$

The tracker keeps a set $\mathcal{O}_u$ of occupied community indices for each user. After every query, it updates the set as $\mathcal{O}_u \leftarrow \mathcal{O}_u \cup \{k^*\}$.

***Query Diversity Score***

The *query diversity score*, also called the *community coverage fraction*, is defined in Equation 6:

$$\tau = \frac{|\mathcal{O}_u|}{K} \in [0, 1]. \tag{6}$$

A benign user who queries one neighborhood touches only a few communities. If the queries fall in $c$ communities, then $\tau = c/K$, which is low when $c \ll K$ and equals $1/K$ in the smallest case. An extraction attacker who samples systematically from all communities drives $\tau \to 1$.

## 3.2 BudgetAllocator

The BudgetAllocator maps $\tau$ to a per-query privacy budget $\varepsilon(\tau)$. A plain sigmoid is a natural choice for this mapping, but it never reaches zero: even at $\tau = 0$, a benign user still receives a budget below $\varepsilon_{\max}$. This adds noise that benign traffic does not need and lowers the utility of the service. The allocator therefore adds a *dead zone*: a range of low $\tau$ values where benign traffic is granted the full budget with no shaping at all.

***Dead-Zone Sigmoid Allocation Function.***

For $\tau$ above the dead-zone threshold $\tau_{\text{dead}}$, the allocator still shapes the budget with a sigmoid, given in Eq. 7:

$$\tilde{g}(\tau) = \frac{1}{1 + e^{-k_{\text{sharp}}(\tau - \tau_0)}}, \tag{7}$$

where $k_{\text{sharp}} > 0$ controls the steepness of the transition and $\tau_0 \in (\tau_{\text{dead}}, 1)$ is the diversity value at which $\tilde{g}(\tau_0) = 0.5$.

***Choosing the Dead-Zone Threshold.***

The value of $\tau_{\text{dead}}$ is not an arbitrary cutoff; it follows from how a benign user's queries land on the graph. A user who studies one part of the graph submits queries whose $L$-hop neighborhoods stay close together, so these queries touch only a small, bounded number of communities even when $K$ is large. If an ordinary session lands in $c$ communities, with $c \ll K$, its diversity score is $\tau = c/K$. Setting $\tau_{\text{dead}}$ slightly above the largest $c/K$ expected from this kind of localized use keeps benign sessions inside the dead zone with room to spare, while still catching a user whose queries begin to spread beyond one or two communities. For example, $\tau_{\text{dead}} = 0.05$ is a reasonable margin when $K \approx 20$ and ordinary sessions touch about one community ($c \approx 1$). This paper treats $\tau_{\text{dead}}$ as a property of the graph's community structure, to be re-tuned per deployment, rather than a fixed constant that transfers unchanged across graphs of very different size.

***Adaptive Budget.***

The per-query budget assigned to user $u$ on query $t$ follows the piecewise rule in Equation 8:

$$\varepsilon(\tau) = \begin{cases} \varepsilon_{\max} & \text{if } \tau \leq \tau_{\text{dead}}, \\ \varepsilon_{\max} + (\varepsilon_0 - \varepsilon_{\max})\, \tilde{g}(\tau) & \text{if } \tau > \tau_{\text{dead}}, \end{cases} \tag{8}$$

with hyperparameters $\varepsilon_0 = 0.01$ (strict privacy, high noise), $\varepsilon_{\max} = 10.0$ (relaxed privacy, low noise), $k_{\text{sharp}} = 20$, $\tau_0 = 0.30$, and $\tau_{\text{dead}} = 0.05$, used throughout the experiments. In Differential Privacy, a higher $\varepsilon(\tau)$ permits more information leakage, and this maps to a lower noise scale. A benign user with $\tau \leq \tau_{\text{dead}}$ therefore receives exactly $\varepsilon(\tau) = \varepsilon_{\max}$, the minimum-noise regime, with no sigmoid shaping applied. An attacker whose diversity exceeds $\tau_{\text{dead}}$ enters the sigmoid region, and as $\tau \to 1$ the budget decreases toward $\varepsilon_0$, the maximum-noise regime. This behavior is intentional. Once the diversity of an attacker is high, the allocator lowers the budget toward the strict baseline. The RDP accountant then exhausts the remaining session budget faster, which triggers the hard cap earlier.

Equation 8 is piecewise defined and has a small jump at $\tau_{\text{dead}}$. Because $\tilde{g}(\tau_{\text{dead}}) \approx 0.0067$, the budget drops from $\varepsilon_{\max} = 10.0$ to approximately 9.93, a decrease of less than 1%. In addition, the sigmoid never reaches its upper limit, so $\varepsilon(1) \approx 0.010008$ and not exactly $\varepsilon_0$. Neither property affects the monotonicity result of Proposition 5 or the accounting guarantee of Section 4. The jump is a decrease, so $\varepsilon(\tau)$ remains non-increasing, and the clamping in Equation 9 depends only on the approved per-query budget and not on the shape of $\varepsilon(\tau)$.

## 3.3 RDPAccountant

The RDPAccountant keeps track of each user's cumulative RDP expenditure $\varepsilon_{\text{cum},u}$ and enforces the hard cap $\varepsilon_{\text{total}}$.

***Budget Check and Clamping.***

Before each query response is issued, the accountant performs the check and update shown in Equations 9 and 10:

$$\varepsilon_q \leftarrow \min(\varepsilon(\tau),\ \varepsilon_{\text{total}} - \varepsilon_{\text{cum},u})\,, \tag{9}$$

$$\varepsilon_{\text{cum},u} \leftarrow \varepsilon_{\text{cum},u} + \varepsilon_q. \tag{10}$$

While a user's diversity score climbs toward $\tau \to 1$ but the session budget has not yet run out, the BudgetAllocator keeps assigning the strict value $\varepsilon_q \to \varepsilon_0$ from Equation 8. For embedding queries this fixes the Gaussian noise scale near its highest *regular* value, given in Equation 11:

$$\sigma_{\max} = \sigma(\varepsilon_0) = \sqrt{\frac{\alpha\, \Delta_2^2}{2\, \varepsilon_0}}, \tag{11}$$

and for label queries this fixes the Randomized Response flip probability near its highest regular value, given in Equation 12:

$$\rho_{\max} = \frac{|\mathcal{C}|}{e^{\varepsilon_0} + |\mathcal{C}| - 1}. \tag{12}$$

With $\varepsilon_0 = 0.01$, $\rho_{\max}$ from Equation 12 is close to 1, so a user who keeps spreading queries across communities receives labels that are almost uniformly random well before the session budget is exhausted. This regime is still governed by a strictly positive per-query budget, $\varepsilon_q = \varepsilon_0 > 0$, and every such query still draws down $\varepsilon_{\text{cum},u}$.

***The Hard Floor: After $\varepsilon_{\text{total}}$ Is Exhausted.***

This regime must not be confused with what happens once the cumulative budget is exhausted. Once $\varepsilon_{\text{cum},u} = \varepsilon_{\text{total}}$, Equation 9 gives $\varepsilon_{\text{total}} - \varepsilon_{\text{cum},u} = 0$, so the clamp forces

$$\varepsilon_q = 0 \tag{13}$$

for every subsequent query in the session, not $\varepsilon_q = \varepsilon_0$. The distinction matters for the paper's cumulative guarantee (Section 4). If every post-exhaustion query were still charged $\varepsilon_0 > 0$, then $\varepsilon_{\text{cum},u}$ would keep growing by $\varepsilon_0$ with every additional query, without limit, and the transcript-wide certificate would fail to hold once the session runs past a few extra queries. Setting the floor to exactly $\varepsilon_q = 0$ is what makes $\varepsilon_{\text{total}}$ a true hard cap: Equation 16 then holds for any session length $T$, since every query issued after exhaustion adds exactly 0 to the sum.

A budget of $\varepsilon_q = 0$ has a different practical consequence for each output type. Substituting $\varepsilon_q = 0$ into Equation 18 gives $\rho(0) = |\mathcal{C}|/(e^0 + |\mathcal{C}| - 1) = |\mathcal{C}|/|\mathcal{C}| = 1$ exactly: Randomized Response returns a label drawn uniformly from $\{1, \ldots, |\mathcal{C}|\}$ with probability 1, so the output no longer depends on $y$ at all, exactly what a mechanism with 0 privacy cost must do. The Gaussian mechanism has no equivalent finite limit, since Equation 24 diverges as $\varepsilon_q \to 0$, so DP-ADAGE cannot sample a noisy embedding at the hard floor. Instead, once $\varepsilon_q = 0$, the API returns a fixed reference vector $\mathbf{e}_{\text{ref}}$ (for example, the all-zero vector, or the dataset's mean clipped embedding, computed once offline and never updated per query), as stated in Equation 14:

$$\tilde{\mathbf{o}} = \begin{cases} \text{one_hot}\big(\text{UniformRandom}(\{1, \ldots, |\mathcal{C}|\})\big) & \text{logits, } \varepsilon_q = 0, \\ \mathbf{e}_{\text{ref}} & \text{embedding/projection, } \varepsilon_q = 0. \end{cases} \tag{14}$$

Because $\mathbf{e}_{\text{ref}}$ is fixed in advance and never depends on the query, returning it costs exactly 0 additional privacy loss, consistent with Equation 13. Algorithm 1 implements this rule directly: the Randomized Response branch already reduces to a pure uniform draw when $\varepsilon_q = 0$, since $\rho_{\text{flip}} = 1$ in that case, and the embedding and projection branches add an explicit check for $\varepsilon_q = 0$ before attempting to sample noise, rather than dividing by zero in Equation 24. Once the session budget runs out, every further response is therefore provably independent of the query, which is what makes the hard cap meaningful in practice: no additional information can be extracted from it,

regardless of which output type the user requests, and regardless of how many more queries the user sends.

### *Per-Query RDP Guarantee*

Every embedding query is answered with the Gaussian mechanism at noise scale $\sigma(\varepsilon_q)$ from Equation 24. Lemma 3.3 confirms that this calibration gives exactly the per-query budget the BudgetAllocator approved, so the accountant's bookkeeping in Equations 9 and 10 is sound. Section 4.3 composes this per-query bound across queries to establish the paper's main cumulative RDP guarantee.

[Per-Query RDP] Let $\varepsilon_q$ be the approved per-query budget from Equation 9, and let $\Delta_2$ be the $\ell_2$-sensitivity of the output function. The Gaussian perturbation with noise scale $\sigma(\varepsilon_q) = \sqrt{\alpha\Delta_2^2/(2\varepsilon_q)}$ from Equation 24 satisfies $(\alpha, \varepsilon_q)$-RDP for every $\alpha > 1$.

*Proof* By the Gaussian mechanism RDP bound [5], adding noise $\mathcal{N}(0, \sigma^2 I)$ to a function with $\ell_2$-sensitivity $\Delta_2$ satisfies $(\alpha, \alpha\Delta_2^2/(2\sigma^2))$-RDP for any noise scale $\sigma$. Setting $\sigma = \sigma(\varepsilon_q)$, so that $\sigma^2 = \alpha\Delta_2^2/(2\varepsilon_q)$, and substituting into the RDP parameter gives Equation 15:

$$\frac{\alpha\Delta_2^2}{2\sigma(\varepsilon_q)^2} = \frac{\alpha\Delta_2^2}{2\left(\dfrac{\alpha\Delta_2^2}{2\varepsilon_q}\right)} = \varepsilon_q. \tag{15}$$

So the mechanism satisfies exactly $(\alpha, \varepsilon_q)$-RDP, for every $\alpha > 1$, which is what the lemma claims. □

Lemma 3.3 covers embedding and projection outputs. Label outputs use Randomized Response instead of the Gaussian mechanism, but as Section 3.4 explains, pure $\varepsilon_q$-DP implies $(\alpha, \varepsilon_q)$-RDP for every $\alpha \geq 1$, so the same per-query budget $\varepsilon_q$ bounds both output types.

### *Composition.*

By RDP composition (Section 2.3), after $T$ queries the cumulative guarantee for user $u$ is given by Equation 16:

$$\sum_{t=1}^{T} \varepsilon_q^{(t)} \;\leq\; \varepsilon_{\text{total}}. \tag{16}$$

The accountant enforces this invariant strictly. The guarantee holds regardless of whether a given query is answered with Randomized Response or with the Gaussian mechanism, since both satisfy $(\alpha, \varepsilon_q)$-RDP for the approved per-query budget, by Lemma 3.3.

### *Choosing the Cumulative Budget.*

The value of $\varepsilon_{\text{total}}$ sets a direct trade-off, and this section states it explicitly rather than leaving it implicit in the experiments. A small $\varepsilon_{\text{total}}$ keeps the composed $(\alpha, \varepsilon_{\text{total}})$-RDP bound tight, but it also pushes benign users into the Noise Floor after only a few queries, which hurts utility. A large $\varepsilon_{\text{total}}$ lets benign users issue many queries before the Noise Floor applies, but the bound on the output likelihood ratio grows with $\varepsilon_{\text{total}}$, so the composed guarantee becomes looser. Past a certain point, the guarantee stays

mathematically correct, but it stops being a meaningful practical bound. As a concrete illustration, the parameters used in this paper's main evaluation ($\varepsilon_{\text{total}} = 200.0$, $\alpha = 8$, $\delta = 10^{-6}$) convert to $\varepsilon^* \approx 201.97$ (Section 4). An $(\varepsilon^*, \delta)$-DP guarantee at that scale allows the output likelihood ratio between adjacent query transcripts to differ by a factor close to $e^{202}$. The certificate is still formally valid, but a bound this loose does not meaningfully limit what a determined attacker can learn from the responses.

***A Formal Criterion for a Meaningful Certificate.***

Equation 16 guarantees that the certificate is *correct* for any $\varepsilon_{\text{total}}$, but the previous paragraph shows it can still be *loose*. Proposition 1 makes precise how small $\varepsilon_{\text{total}}$ must be for the final certificate to bound the attacker's advantage by a chosen factor, rather than only being a valid but uninformative number.

**Proposition 1** (Meaningful-Budget Threshold) *Fix $\alpha > 1$ and $\delta \in (0, 1)$, and let $M > 1$ be a likelihood-ratio bound the deployer is willing to treat as protective (for example, $M = e^{10}$, a range common in production differential-privacy deployments). The converted certificate $\varepsilon^* = \varepsilon_{\text{total}} + \log(1/\delta)/(\alpha - 1)$ satisfies $e^{\varepsilon^*} \leq M$ if and only if Equation 17 holds:*

$$\varepsilon_{\text{total}} \;\leq\; \log M - \frac{\log(1/\delta)}{\alpha - 1}. \tag{17}$$

*Proof* The condition $e^{\varepsilon^*} \leq M$ is equivalent to $\varepsilon^* \leq \log M$. Substituting $\varepsilon^* = \varepsilon_{\text{total}} + \log(1/\delta)/(\alpha - 1)$ and rearranging for $\varepsilon_{\text{total}}$ gives Equation 17 directly. □

With $\alpha = 8$ and $\delta = 10^{-6}$, the second term in Equation 17 equals $\log(10^6)/7 \approx 1.97$. Even a generous target of $M = e^{10}$ then permits only $\varepsilon_{\text{total}} \lesssim 8.0$, an order of magnitude below the 200.0 configuration used in this paper's main results. Section 5's budget-cap ablation already covers this tighter regime ($\varepsilon_{\text{total}} \in \{1.0, 10.0\}$) and shows it costs benign users almost nothing in downstream accuracy. This paper therefore reports $\varepsilon_{\text{total}} = 200.0$ as a *high-availability* configuration, chosen to stress-test the mechanism under a generous budget, and recommends a single-digit $\varepsilon_{\text{total}}$, per Proposition 1, for any deployment that needs the formal certificate itself, and not only the empirical accuracy numbers, to represent a meaningful bound on leakage.

## 3.4 Perturbation Mechanism

The perturbation mechanism used for a query depends on the output type the API returns. Discrete probability outputs (Setup A) use Randomized Response on the predicted class label. Continuous embedding outputs (Setup B) and their downstream projections (Setup C) keep the Gaussian mechanism. Gaussian noise is a natural fit for high-dimensional continuous vectors, but it destroys the signal in a low-dimensional discrete label.

***Randomized Response for Label Outputs.***

Let $y = \arg\max\big(\text{softmax}(\mathbf{z}_v)\big)$ be the class the victim model predicts for node $v$. Given the approved per-query budget $\varepsilon_q$, the API computes a flip probability using

Equation 18:

$$\rho(\varepsilon_q) = \frac{|\mathcal{C}|}{e^{\varepsilon_q} + |\mathcal{C}| - 1}, \tag{18}$$

and returns a perturbed label according to Equation 19:

$$\tilde{y} = \begin{cases} y & \text{with probability } 1 - \rho(\varepsilon_q), \\ \text{drawn uniformly from } \{1, \ldots, |\mathcal{C}|\} & \text{with probability } \rho(\varepsilon_q), \end{cases} \tag{19}$$

with the API returning $\tilde{\mathbf{p}}_v = \text{one_hot}(\tilde{y})$.

***Exact Privacy Guarantee of Randomized Response***

Equation 19 gives each possible output an exact probability, so the privacy guarantee can be read off directly rather than only bounded. For the true label $y$, the mechanism returns $y$ either because no flip happens or because the uniform draw happens to land back on $y$:

$$\Pr[\tilde{y} = y] = \big(1 - \rho(\varepsilon_q)\big) + \frac{\rho(\varepsilon_q)}{|\mathcal{C}|}. \tag{20}$$

For any other label $y' \neq y$, the mechanism can only reach $y'$ through the uniform draw:

$$\Pr[\tilde{y} = y'] = \frac{\rho(\varepsilon_q)}{|\mathcal{C}|}. \tag{21}$$

Substituting $\rho(\varepsilon_q) = |\mathcal{C}|/(e^{\varepsilon_q} + |\mathcal{C}| - 1)$ from Equation 18 into Equations 20 and 21 and simplifying gives the exact output distribution in Equation 22:

$$\Pr[\tilde{y} = y] = \frac{e^{\varepsilon_q}}{e^{\varepsilon_q} + |\mathcal{C}| - 1}, \qquad \Pr[\tilde{y} = y'] = \frac{1}{e^{\varepsilon_q} + |\mathcal{C}| - 1}. \tag{22}$$

Taking the ratio of the two terms in Equation 22 gives Equation 23:

$$\frac{\Pr[\tilde{y} = y]}{\Pr[\tilde{y} = y']} = e^{\varepsilon_q}. \tag{23}$$

This is the worst-case likelihood ratio over every pair of possible true labels, since Equation 21 does not depend on which label is queried, so no other pair of outputs can produce a larger ratio. Equation 23 shows that Randomized Response with the flip probability from Equation 18 satisfies pure $\varepsilon_q$-DP exactly, not only as an upper bound: the maximum output likelihood ratio between any two adjacent labels equals $e^{\varepsilon_q}$ exactly. Pure $\varepsilon$-DP implies $(\alpha, \varepsilon)$-RDP for every $\alpha \geq 1$ (Mironov, 2017), so the RDP accounting in Section 3.3 applies to Randomized Response outputs without change. Section 4 places this per-query result inside the paper's overall RDP composition and the final RDP-to-DP conversion.

***Embedding Perturbation.***

For node embeddings $\mathbf{e}_v \in \mathbb{R}^d$, the API enforces an output clipping norm $B$, so that $\|\mathbf{e}_v\|_2 \leq B$. The sensitivity of the clipped embedding is $\Delta_2 = 2B$, and the noise scale follows the standard RDP Gaussian calibration in Equation 24:

$$\sigma(\varepsilon_q) = \sqrt{\frac{\alpha\,\Delta_2^2}{2\,\varepsilon_q}}. \tag{24}$$

This paper writes the noise scale as $\sigma(\varepsilon_q)$ throughout, since it is a function of the per-query budget $\varepsilon_q$ approved by the RDPAccountant (Section 3.3), not of the diversity score $\tau$ directly. $\tau$ only enters indirectly, through Equation 8. The perturbed embedding is given by Equation 25:

$$\tilde{\mathbf{e}}_v = \mathrm{clip}(\mathbf{e}_v, B) + \boldsymbol{\eta}, \quad \boldsymbol{\eta} \sim \mathcal{N}\big(\mathbf{0},\, \sigma(\varepsilon_q)^2\, \mathbf{I}_d\big). \tag{25}$$

***Post-Processing Invariance.***

Any deterministic post-processing of a DP output is also DP, by the post-processing theorem [11]. So for projection outputs (Setup C), the API perturbs the embedding first with Equation 25 and then applies the deterministic t-SNE transform $\gamma = H(\tilde{\mathbf{e}}_v)$, which inherits the same $(\varepsilon_q, \delta)$-DP guarantee as the perturbed embedding it is computed from.

***Complete Query-Response Protocol.***

Algorithm 1 ties the four components above into a single query-response protocol. It combines the community lookup of Equation 5, the dead-zone budget rule of Equation 8, the clamp-and-accumulate step of Equations 9 and 10, and the output-specific perturbation of Equations 18–19 and Equations 24–25 into one procedure that runs on every incoming query.

## 3.5 Utility Analysis

The BudgetAllocator is built to keep benign users and extraction attackers in two separate noise regimes. This section examines both regimes and shows how far apart they end up.

A benign user who queries within one neighborhood keeps a low diversity score, so $\tau \leq \tau_{\text{dead}}$. Under the dead-zone rule in Equation 8, this user receives the relaxed budget $\varepsilon(\tau) = \varepsilon_{\max}$, with no sigmoid shaping at all. By Equation 24, the embedding noise scale for this user is given in Equation 26:

$$\sigma_{\text{benign}} = \sigma(\varepsilon_{\max}) = \sqrt{\frac{\alpha\,\Delta_2^2}{2\,\varepsilon_{\max}}}. \tag{26}$$

Since $\varepsilon_{\max} = 10.0$ is the largest budget the allocator ever grants, $\sigma_{\text{benign}}$ is the smallest noise scale in the system. This is the minimum-noise regime, and it is what preserves downstream accuracy for normal traffic.

**Algorithm 1** DP-ADAGE Query-Response Protocol

**Require:** Victim model $f_\theta$, training graph $G$, configuration $(\varepsilon_0, \varepsilon_{\max}, \varepsilon_{\text{total}}, k_{\text{sharp}}, \tau_0, \tau_{\text{dead}}, \alpha, \delta, B)$, fixed reference vector $e_{\text{ref}}$

**Ensure:** Perturbed API response $\tilde{o}$

**Offline (once):**
$\{C_k\}_{k=1}^{K} \leftarrow \text{Louvain}(G)$
$\omega_k \leftarrow \frac{1}{|C_k|} \sum_{v \in C_k} f_\theta^{(L-1)}(v), \quad \forall k$
**Per-query** $(u, \text{node } v, \text{output type } \rho)$:
$e \leftarrow f_\theta^{(L-1)}(v)$
$k^* \leftarrow \arg\min_k \|e - \omega_k\|_2$
$O_u \leftarrow O_u \cup \{k^*\}$
$\tau \leftarrow |O_u|/K$
**if** $\tau \le \tau_{\text{dead}}$ **then**
    $\varepsilon(\tau) \leftarrow \varepsilon_{\max}$
**else**
    $\varepsilon(\tau) \leftarrow \varepsilon_{\max} + \frac{\varepsilon_0 - \varepsilon_{\max}}{1 + e^{-k_{\text{sharp}}(\tau - \tau_0)}}$
**end if**
$\varepsilon_q \leftarrow \min(\varepsilon(\tau), \varepsilon_{\text{total}} - \varepsilon_{\text{cum},u})$
$\varepsilon_{\text{cum},u} \leftarrow \varepsilon_{\text{cum},u} + \varepsilon_q$
**if** $\rho = \text{logits}$ **then**
    $y \leftarrow \arg\max(\text{softmax}(f_\theta^{(L)}(v)))$
    $\rho_{\text{flip}} \leftarrow \frac{|C|}{e^{\varepsilon_q} + |C| - 1}$
    **if** $\text{rand}() < 1 - \rho_{\text{flip}}$ **then**
        $\tilde{o} \leftarrow \text{one_hot}(y)$
    **else**
        $\tilde{o} \leftarrow \text{one_hot}(\text{UniformRandom}(\{1, \ldots, |C|\}))$
    **end if**
**else if** $\rho = \text{embedding}$ **then**
    **if** $\varepsilon_q = 0$ **then**
        $\tilde{o} \leftarrow e_{\text{ref}}$ ▷ hard floor
    **else**
        $\sigma(\varepsilon_q) \leftarrow \sqrt{\alpha(2B)^2/(2\varepsilon_q)}$
        $\tilde{o} \leftarrow \text{clip}(e, B) + \mathcal{N}(0, \sigma(\varepsilon_q)^2 I)$
    **end if**
**else if** $\rho = \text{projection}$ **then**
    **if** $\varepsilon_q = 0$ **then**
        $\tilde{o} \leftarrow H(e_{\text{ref}})$ ▷ hard floor
    **else**
        $\sigma(\varepsilon_q) \leftarrow \sqrt{\alpha(2B)^2/(2\varepsilon_q)}$
        $\tilde{e} \leftarrow \text{clip}(e, B) + \mathcal{N}(0, \sigma(\varepsilon_q)^2 I)$
        $\tilde{o} \leftarrow H(\tilde{e})$
    **end if**
**end if**
**return** $\tilde{o}$

An extraction attacker who spreads queries across many communities drives $\tau \to 1$. As Section 3.2 explains, the sigmoid term then pushes the budget down toward the strict baseline $\varepsilon_0$, so while the session budget still has room left, the attacker's queries are answered at $\varepsilon_q \to \varepsilon_0$, giving the noise scale already defined in Equation 11:

$$\sigma_{\text{attacker}} = \sigma(\varepsilon_0) = \sqrt{\frac{\alpha\,\Delta_2^2}{2\,\varepsilon_0}}. \tag{27}$$

With $\varepsilon_0 = 0.01$, this is the maximum *regular*-noise regime, and it applies to every query an attacker sends once their diversity score crosses $\tau_0$, for as long as the session budget lasts. If the attacker keeps querying past that point until the RDPAccountant's hard cap is reached (Section 3.3), later queries fall to the stricter $\varepsilon_q = 0$ floor described there, which returns an output that no longer depends on the query at all, rather than the finite noise scale $\sigma_{\text{attacker}}$ above.

The size of the gap between the two regimes follows directly from Equations 26 and 27. Taking the ratio of the two noise variances gives Equation 28:

$$\frac{\sigma^2_{\text{attacker}}}{\sigma^2_{\text{benign}}} = \frac{\alpha\Delta_2^2/(2\varepsilon_0)}{\alpha\Delta_2^2/(2\varepsilon_{\max})} = \frac{\varepsilon_{\max}}{\varepsilon_0}. \tag{28}$$

With the hyperparameters used throughout the experiments, $\varepsilon_{\max} = 10.0$ and $\varepsilon_0 = 0.01$, this ratio equals 1000: an attacker's embedding queries carry a thousand times more noise variance than a benign user's queries. Randomized Response label queries show the same pattern, since $\rho(\varepsilon_q)$ in Equation 18 is a decreasing function of $\varepsilon_q$, so the flip probability is lowest for benign users at $\varepsilon_{\max}$ and highest for attackers pushed toward $\varepsilon_0$.

This separation is a property of $\varepsilon_{\max}$ and $\varepsilon_0$ alone, so it does not depend on $\varepsilon_{\text{total}}$. Tightening the total budget, as Proposition 1 recommends for a meaningful certificate, changes how quickly a user reaches the Noise Floor, but it does not narrow the gap between the benign and attacker noise regimes while a session is still active.

This gap is the mechanism behind the utility–security trade-off that motivates DP-ADAGE. A static, query-independent noise level has to pick one point on this scale for every user, which either leaks too much to attackers or destroys accuracy for benign traffic. The dead-zone allocator instead keeps benign users near $\sigma_{\text{benign}}$, close to noise-free, while pushing attackers toward $\sigma_{\text{attacker}}$, close to the Noise Floor from Section 3.3, using the same query diversity signal that the CommunityTracker already computes.

# 4 Formal Privacy Analysis

This section establishes the formal privacy guarantees of DP-ADAGE. The proofs build on the Gaussian-mechanism RDP bound and adaptive composition [5], together with the RDP-to-DP conversion theorem [13].

## 4.1 Per-Query RDP Guarantee

**Theorem 2** (Per-Query RDP) *Let $\varepsilon_q$ be the approved per-query budget (Eq. (9)) and let $\Delta_2$ be the $\ell_2$-sensitivity of the output function. The Gaussian perturbation with noise scale $\sigma_\tau = \sqrt{\alpha\Delta_2^2/(2\varepsilon_q)}$ (Eq. (24)) satisfies $(\alpha, \varepsilon_q)$-RDP for all $\alpha > 1$.*

*Proof* By the Gaussian mechanism RDP bound [5], the Gaussian mechanism with noise variance $\sigma^2$ applied to a function with $\ell_2$-sensitivity $\Delta_2$ satisfies $\left(\alpha, \frac{\alpha\Delta_2^2}{2\sigma^2}\right)$-RDP. Substituting $\sigma_\tau^2 = \alpha\Delta_2^2/(2\varepsilon_q)$ gives RDP order $\alpha$ with parameter $\frac{\alpha\Delta_2^2}{2\cdot\alpha\Delta_2^2/(2\varepsilon_q)} = \varepsilon_q$. □

## 4.2 Sensitivity Analysis

***Logit Sensitivity.***

For logit outputs $\mathbf{z}_v \in \mathbb{R}^{|\mathcal{C}|}$ generated by a network with gradient clipping norm $C$, the $\ell_2$-sensitivity satisfies $\Delta_2 \leq 2C$ because any two adjacent datasets (differing in one node's features) change the logit vector by at most $2C$ in $\ell_2$ norm [12].

***Embedding Sensitivity.***

For node embeddings $\mathbf{e}_v \in \mathbb{R}^d$ with $\|\mathbf{e}_v\|_2 \leq B$ for all $v$, removing or adding a single node shifts the embedding by at most $\Delta_2 = 2B$, giving the bound used in Eq. (24).

## 4.3 Cumulative Privacy via RDP Composition

**Theorem 3** (Cumulative RDP) *Let user $u$ issue $T$ queries receiving approved budgets $\varepsilon_q^{(1)}, \ldots, \varepsilon_q^{(T)}$. The adaptive composition of the $T$ corresponding Gaussian mechanisms satisfies $(\alpha, \sum_{t=1}^{T} \varepsilon_q^{(t)})$-RDP.*

*Proof* Direct application of the RDP composition lemma [5]: the RDP parameter of the composition of mechanisms $\mathcal{M}_1, \ldots, \mathcal{M}_T$, each satisfying $(\alpha, \varepsilon_q^{(t)})$-RDP, is at most $\sum_{t=1}^{T} \varepsilon_q^{(t)}$. The RDPAccountant clamping (Eq. (9)) ensures $\sum_{t=1}^{T} \varepsilon_q^{(t)} \leq \varepsilon_{\text{total}}$. □

[Hard Cap] For any user $u$ and any number of queries $T$, the total RDP expenditure satisfies:

$$\sum_{t=1}^{T} \varepsilon_q^{(t)} \;\leq\; \varepsilon_{\text{total}}. \tag{29}$$

Once $\varepsilon_{\text{cum},u} = \varepsilon_{\text{total}}$, all subsequent responses carry maximum noise; additional queries yield zero incremental information gain in the RDP sense.

## 4.4 Final $(\varepsilon, \delta)$-DP Certificate

**Theorem 4** ($(\varepsilon, \delta)$-DP Guarantee) *For any $\delta \in (0, 1)$, the full DP-ADAGE interaction transcript for user $u$ (all $T$ query responses) satisfies $(\varepsilon^*, \delta)$-DP where:*

$$\varepsilon^* = \varepsilon_{\text{total}} + \frac{\log(1/\delta)}{\alpha - 1}. \tag{30}$$

*Proof* Theorem 3 and Corollary 4.3 establish $(\alpha, \varepsilon_{\text{total}})$-RDP for the transcript. Applying the conversion theorem [13]: any $(\alpha, \varepsilon_{\text{RDP}})$-RDP mechanism is $(\varepsilon_{\text{RDP}} + \log(1/\delta)/(\alpha - 1), \delta)$-DP. Substituting $\varepsilon_{\text{RDP}} = \varepsilon_{\text{total}}$ yields Eq. (30). □

***Concrete Parameters.***

With $\alpha = 8$, $\varepsilon_{\text{total}} = 200.0$, and $\delta = 10^{-6}$, Theorem 4 certifies $(200.0 + \log(10^6)/7,\ 10^{-6})$-DP $= (201.97,\ 10^{-6})$-DP for the full interaction, regardless of the number of queries issued.

This certificate bounds transcript leakage under the stated assumptions. It does not by itself guarantee protection against model extraction (see Section 6.2).

## 4.5 Utility Analysis

For a benign user with community coverage $\tau \ll \tau_0$, the sigmoid $g(\tau) \approx 0$, so $\varepsilon(\tau) \approx \varepsilon_{\max}$. Benign users therefore receive the largest per-query budget, which means the smallest noise. Inside the dead zone ($\tau \leq \tau_{\text{dead}}$), the budget is exactly $\varepsilon_{\max}$. The noise scale reduces to:

$$\sigma_\tau^{\text{benign}} \approx \sqrt{\frac{\alpha\, \Delta_2^2}{2\varepsilon_{\max}}}. \tag{31}$$

This scale is small whenever $\varepsilon_{\max}$ is large. With $\alpha = 8$, $\Delta_2 = 2C = 2$, and $\varepsilon_{\max} = 10.0$, it is $\sigma_\tau^{\text{benign}} \approx 1.26$. A user at the strictest budget $\varepsilon_0 = 0.01$ would face $\sigma_\tau = 40$, which is $\sqrt{\varepsilon_{\max}/\varepsilon_0} \approx 31.6$ times larger. The allocator thus keeps the noise low for users who stay in a small part of the graph, and it raises the noise only as the diversity score $\tau$ grows. The experiments agree with this design. Averaged over the five datasets, DP-ADAGE reaches a benign accuracy of $80.5 \pm 0.4\%$, within 0.1 points of the undefended model (see Section 5). This design separates the noise regimes for benign and adversarial users, which addresses the utility–security tension of static-noise defenses.

This benefit lasts only while the budget lasts. A benign user who is charged $\varepsilon_{\max} = 10.0$ per query spends a cap of $\varepsilon_{\text{total}} = 200.0$ after about 20 queries. After that point, the response no longer depends on the query. The utility result above therefore describes accuracy under a fresh budget, not the experience of a high-volume user in a long session.

## 4.6 Monotonicity and Fairness

**Proposition 5** (Monotone Noise) *$\varepsilon(\tau)$ is non-increasing in $\tau$, and strictly decreasing for $\tau > \tau_{dead}$. Consequently, the noise scale $\sigma_\tau$ is non-decreasing in $\tau$: users who probe more graph communities receive responses with* more *noise.*

*Proof* The sigmoid $\tilde{g}(\tau)$ in Eq. (7) is strictly increasing for $k_{\text{sharp}} > 0$. Since $\varepsilon_0 < \varepsilon_{\max}$, the factor $(\varepsilon_0 - \varepsilon_{\max})$ in Eq. (8) is negative, so $\varepsilon(\tau)$ is strictly decreasing for $\tau > \tau_{\text{dead}}$ and constant below it. From Eq. (24), $\sigma_\tau \propto \varepsilon_q^{-1/2}$ is strictly decreasing in $\varepsilon_q$, so $\sigma_\tau$ is non-decreasing in $\tau$. □

This monotonicity property ensures that the defense is *fair*: the noise injected is proportional to the information hazard of the query, not arbitrarily applied to all users.

# 5 Evaluation

This section reports the results of DP-ADAGE on five citation-network datasets: Cora, Citeseer, Pubmed, ACM, and DBLP. Each dataset is tested under six conditions: NoDefense, StaticNoise, ADAGE, PRADA, DAWN, and DP-ADAGE. Every condition was run with five random seeds (42 to 46). Unless a table says otherwise, values are the mean $\pm$ the sample standard deviation (with $n-1$ in the denominator) over these five seeds. For metrics that are averaged over datasets, the average is taken within each seed first, and the standard deviation is then computed across seeds.

For every condition, Table 2 reports five metrics. These are the benign accuracy of the downstream classifier, the surrogate accuracy that an attacker reaches through model extraction, the attacker's fidelity to the victim model, the total RDP budget spent by the RDPAccountant, and the final query diversity score $\tau$ of the querying user. The section then covers the query-budget trajectories, an ablation of the budget cap, the utility–security trade-off, a significance test, a test against a budget-aware attacker, a scalability run on Flickr, and the cost in latency, memory, and throughput. It ends with a note on what the evidence can and cannot support.

## 5.1 Experimental Setup

Table 1 lists the configuration used in the experiments. The victim model is a two-layer GCN with 64 hidden units. It is trained for 200 epochs with a learning rate of 0.01, a weight decay of $5 \times 10^{-4}$, and a dropout rate of 0.5. The surrogate model has the same GCN architecture. Each graph is split into training, validation, and test nodes with a ratio of 80/10/10. The API returns the predicted label (the `logits` output type). The sweep over privacy budgets uses five datasets: ACM, Citeseer, Cora, DBLP, and Pubmed.

The CommunityTracker divides each graph into communities with the Louvain algorithm. The number of communities $K$ is not fixed in advance. It is set by the algorithm, so it depends on the graph and on the seed. The query diversity score $\tau$ is the number of communities the user has queried, divided by $K$. This is consistent with the logged values of $\tau$, which are exact fractions with a denominator that stays the same within one run. The denominators are 101–106 on Cora, 467–473 on Citeseer, 36–45 on Pubmed, 80–82 on ACM, and 618–623 on DBLP. Because $K$ differs so much between graphs, one new community moves $\tau$ by very different amounts: about 0.2% on Citeseer and DBLP, 1% on Cora and ACM, and 2–3% on Pubmed. This is one reason why the budget runs out at different speeds on different datasets.

The attacker uses uncertainty sampling. The main trajectories run to 200 queries. The Cora case study in Section 5.6 stops at 100 queries, the Flickr run in Section 5.11 also stops at 100, and the adaptive-attacker test in Section 5.10 extends the horizon to 500 queries.

The legitimate-user evaluation in Section 5.4 uses a benign query policy. Each benign session sends 200 sequential queries to the API. Two protocols are compared.

In the session protocol, the budget accumulates over all queries of one user. In the fresh protocol, the budget does not accumulate across queries. The evaluation reports the query at which the budget reaches its floor, the rolling accuracy over a window of 10 queries, the accuracy after 50 and 100 queries, and the utility retention (the normalized area under the accuracy curve). It also reports two availability measures. Transport availability is the fraction of queries that receive any response. Informative availability is the fraction of queries that receive a response before the budget reaches its floor.

The per-query budget ranges from $\varepsilon_0 = 0.01$ for the strictest case to $\varepsilon_{\max} = 10.0$ for the most relaxed case. Both values can be read from the result files. A user with a low diversity score is charged 10.0 units per query (99.5 units after 10 queries on Cora), and on Pubmed the charge of users with a high score settles at exactly 0.010 units per query. The allocator is a sigmoid with steepness $k_{\text{sharp}} = 20$ and midpoint $\tau_0 = 0.30$. A dead zone below a threshold $\tau_{\text{dead}}$ gives the full budget of 10.0. The logged budgets give bounds for this threshold, because every interval with $\tau \leq 0.021$ was charged exactly 10.0 per query, and every interval that ended at $\tau \geq 0.078$ was charged less. The threshold therefore lies between 0.02 and 0.08.

The privacy evaluation uses four total budgets, $\varepsilon_{\text{total}} \in \{1.0, 5.0, 10.0, 200.0\}$, with RDP order $\alpha = 8$ and $\delta = 10^{-6}$. The value 200.0 is the reference setting of the main attacker–defense results. Its cap is visible in the result files, where the cumulative budget stops at exactly 200.0. The three smaller values were added to show the trade-offs among privacy, detection effectiveness, and legitimate-user accuracy more clearly. By Theorem 4.4, each cap gives a transcript-level guarantee of $\varepsilon^* = \varepsilon_{\text{total}} + \log(10^6)/7$. The resulting values are approximately 2.97, 6.97, 11.97, and 201.97, respectively.

The threshold of about 8.0 that Proposition 3.2 gives for a likelihood-ratio bound of $e^{10}$ separates these cases. The caps of 1.0 and 5.0 fall below it, so their certificates are meaningful. The cap of 10.0 falls above it. The cap of 200.0 is far above it, so the reference results describe a high-availability setting. They do not describe a certificate that is meaningful in practice. Section 5.7 reports the sweep over all four caps. Both clipping norms, $C$ for logits and $B$ for embeddings, are set to 1.0.

## 5.2 Main Results

Table 2 compares all six defenses on each dataset. Attacker fidelity measures how closely the surrogate's predictions match the victim's predictions rather than the true labels. It is therefore a stricter test of extraction success than surrogate accuracy alone.

On all five datasets, DP-ADAGE keeps benign accuracy within 0.1 percentage points of the undefended baseline. This gap is much smaller than the seed-to-seed standard deviation, which is between 0.2 and 1.8 points. A legitimate user therefore sees almost no drop in service quality. StaticNoise and PRADA also preserve benign accuracy. DAWN is the outlier. It costs between 2.4 and 3.0 points of benign accuracy on every dataset. It also gives little protection. Its surrogate accuracy equals the undefended value on Cora (80.5%) and is lower by 0.6 to 4.4 points on the other four datasets. Most of these gaps are within one standard deviation. DAWN is therefore the only defense that gives up utility without a clear security gain.

**Table 1** Configuration used in the experiments. The total budget is 200.0 in the reference setting and 1.0, 5.0, 10.0, or 200.0 in the budget-cap sweep. The number of communities $K$ is set by the Louvain algorithm for each graph and seed.

| Group | Parameter | Value |
|---|---|---|
| Defense | Minimum per-query budget $\varepsilon_0$ | 0.01 |
| | Maximum per-query budget $\varepsilon_{\max}$ | 10.0 |
| | Total budget $\varepsilon_{\text{total}}$ | 200.0 (reference); 1.0, 5.0, 10.0, 200.0 (sweep) |
| | Sigmoid steepness $k_{\text{sharp}}$ | 20 |
| | Sigmoid midpoint $\tau_0$ | 0.30 |
| | Dead-zone threshold $\tau_{\text{dead}}$ | 0.05 |
| | RDP order $\alpha$ | 8 |
| | Failure probability $\delta$ | $10^{-6}$ |
| Clipping and output | Logit clipping norm $C$ | 1.0 |
| | Embedding clipping norm $B$ | 1.0 |
| | Output type | logits (labels) |
| Communities | Method | Louvain |
| | Number of communities $K$ | Dynamic (set by Louvain, e.g., $K = 102$ for Cora, seed 42) |
| Victim model | Architecture | GCN, 64 hidden units |
| | Epochs | 200 |
| | Learning rate | 0.01 |
| | Weight decay | $5 \times 10^{-4}$ |
| | Dropout | 0.5 |
| Attacker and users | Surrogate model | GCN, 64 hidden units |
| | Query strategy | uncertainty sampling |
| | Maximum queries (attacker) | 100 (Cora case study, Flickr), 200 (main), 500 (adaptive) |
| | Queries per legitimate session | 200 (session and fresh protocols) |
| Data | Datasets (budget sweep) | ACM, Citeseer, Cora, DBLP, Pubmed |
| | Train/val/test split | 0.8/0.1/0.1 |
| | Runs per configuration | 5 (seeds 42–46); 1 for Flickr, latency, and profiling |

DP-ADAGE gives the lowest attacker accuracy and fidelity on every dataset. On Cora, attacker surrogate accuracy falls from 80.5% under NoDefense to 26.2% under DP-ADAGE. Fidelity drops from 91.5% to 26.1%. ADAGE and PRADA also reduce attacker accuracy on Cora, but only to 70.7% and 65.3%. The pattern differs across the other datasets. PRADA is the second-best defense on Citeseer (59.1%), ACM (70.7%), and DBLP (71.0%). ADAGE is second on Pubmed (54.9%), where PRADA is weak at 71.4%. ADAGE has almost no effect on Citeseer (70.5% against 71.4%) and DBLP (82.3% against 81.9%). StaticNoise stays close to the undefended numbers everywhere. DAWN does the same on all datasets except ACM, where it reaches 82.1% against 86.5% for NoDefense. The standard deviation of DAWN on ACM is 4.6 points, so even this gap is small.

Pubmed is the dataset where DP-ADAGE protects least. The residual attacker accuracy of $49.0 \pm 8.0\%$ and fidelity of $52.3 \pm 8.4\%$ are well above the values on the other datasets. It is also the only dataset where the RDPAccountant does not always use the full budget. The mean spend is $162.9 \pm 34.0$ of 200.0 units. Only one of the five seeds reaches the cap, and the final query diversity of the other four stays high

**Table 2** Per-dataset benign accuracy, attacker (surrogate) accuracy, attacker fidelity, total RDP budget consumed, and final query diversity score $\tau$ at query 200 ($\varepsilon_{\text{total}} = 200.0$). Values are the mean ± standard deviation over five seeds. Bold marks the DP-ADAGE row for each dataset. Budget is N/A where no RDP accounting applies. The $\tau$ of NoDefense, StaticNoise, PRADA, and DAWN is not defined and is shown as 0.000.

| Dataset | Defense | Benign Acc. (%) | Att. Surr. Acc. (%) | Att. Fid. (%) | Budget | $\tau$ |
|---|---|---|---|---|---|---|
| Cora | NoDefense | $80.6 \pm 0.6$ | $80.5 \pm 0.7$ | $91.5 \pm 0.5$ | N/A | 0.000 |
| | StaticNoise | $80.5 \pm 0.5$ | $80.6 \pm 0.9$ | $91.7 \pm 0.8$ | N/A | 0.000 |
| | ADAGE | $80.6 \pm 0.6$ | $70.7 \pm 2.8$ | $77.2 \pm 2.4$ | N/A | $0.626 \pm 0.056$ |
| | PRADA | $80.6 \pm 0.6$ | $65.3 \pm 2.2$ | $71.6 \pm 2.7$ | N/A | 0.000 |
| | DAWN | $77.7 \pm 0.3$ | $80.5 \pm 0.9$ | $90.0 \pm 0.7$ | N/A | 0.000 |
| | **DP-ADAGE** | $\mathbf{80.6 \pm 0.6}$ | $\mathbf{26.2 \pm 4.2}$ | $\mathbf{26.1 \pm 4.4}$ | $\mathbf{200.0 \pm 0.0}$ | $\mathbf{0.656 \pm 0.020}$ |
| Citeseer | NoDefense | $71.3 \pm 0.3$ | $71.4 \pm 0.7$ | $87.6 \pm 1.1$ | N/A | 0.000 |
| | StaticNoise | $71.2 \pm 0.4$ | $71.3 \pm 1.1$ | $86.7 \pm 1.7$ | N/A | 0.000 |
| | ADAGE | $71.3 \pm 0.3$ | $70.5 \pm 0.4$ | $87.2 \pm 0.5$ | N/A | $0.303 \pm 0.007$ |
| | PRADA | $71.3 \pm 0.3$ | $59.1 \pm 6.1$ | $65.7 \pm 6.3$ | N/A | 0.000 |
| | DAWN | $68.3 \pm 1.1$ | $69.4 \pm 2.3$ | $85.0 \pm 2.5$ | N/A | 0.000 |
| | **DP-ADAGE** | $\mathbf{71.2 \pm 0.3}$ | $\mathbf{26.3 \pm 5.9}$ | $\mathbf{27.0 \pm 5.8}$ | $\mathbf{200.0 \pm 0.0}$ | $\mathbf{0.301 \pm 0.006}$ |
| Pubmed | NoDefense | $79.4 \pm 0.2$ | $79.2 \pm 0.4$ | $93.8 \pm 0.6$ | N/A | 0.000 |
| | StaticNoise | $79.3 \pm 0.4$ | $78.8 \pm 0.5$ | $94.3 \pm 0.4$ | N/A | 0.000 |
| | ADAGE | $79.4 \pm 0.2$ | $54.9 \pm 5.7$ | $57.9 \pm 7.9$ | N/A | $0.944 \pm 0.046$ |
| | PRADA | $79.4 \pm 0.2$ | $71.4 \pm 4.6$ | $82.9 \pm 6.0$ | N/A | 0.000 |
| | DAWN | $77.0 \pm 0.3$ | $78.5 \pm 0.9$ | $93.1 \pm 1.4$ | N/A | 0.000 |
| | **DP-ADAGE** | $\mathbf{79.4 \pm 0.2}$ | $\mathbf{49.0 \pm 8.0}$ | $\mathbf{52.3 \pm 8.4}$ | $\mathbf{162.9 \pm 34.0}$ | $\mathbf{0.928 \pm 0.034}$ |
| ACM | NoDefense | $87.1 \pm 1.7$ | $86.5 \pm 1.1$ | $98.1 \pm 1.1$ | N/A | 0.000 |
| | StaticNoise | $87.1 \pm 1.5$ | $86.4 \pm 1.3$ | $98.1 \pm 0.8$ | N/A | 0.000 |
| | ADAGE | $87.1 \pm 1.7$ | $75.8 \pm 2.5$ | $82.8 \pm 4.3$ | N/A | $0.663 \pm 0.060$ |
| | PRADA | $87.1 \pm 1.7$ | $70.7 \pm 2.9$ | $77.3 \pm 2.0$ | N/A | 0.000 |
| | DAWN | $84.1 \pm 2.1$ | $82.1 \pm 4.6$ | $93.3 \pm 4.6$ | N/A | 0.000 |
| | **DP-ADAGE** | $\mathbf{87.0 \pm 1.8}$ | $\mathbf{33.3 \pm 3.4}$ | $\mathbf{34.0 \pm 3.7}$ | $\mathbf{200.0 \pm 0.0}$ | $\mathbf{0.712 \pm 0.043}$ |
| DBLP | NoDefense | $84.5 \pm 0.9$ | $81.9 \pm 1.3$ | $93.2 \pm 0.8$ | N/A | 0.000 |
| | StaticNoise | $84.5 \pm 0.9$ | $81.9 \pm 0.5$ | $93.3 \pm 0.9$ | N/A | 0.000 |
| | ADAGE | $84.5 \pm 0.9$ | $82.3 \pm 0.8$ | $93.4 \pm 0.7$ | N/A | $0.193 \pm 0.009$ |
| | PRADA | $84.5 \pm 0.9$ | $71.0 \pm 2.0$ | $77.4 \pm 2.0$ | N/A | 0.000 |
| | DAWN | $81.5 \pm 0.9$ | $80.8 \pm 2.2$ | $90.7 \pm 3.3$ | N/A | 0.000 |
| | **DP-ADAGE** | $\mathbf{84.5 \pm 0.9}$ | $\mathbf{37.8 \pm 7.4}$ | $\mathbf{38.3 \pm 7.9}$ | $\mathbf{200.0 \pm 0.0}$ | $\mathbf{0.223 \pm 0.009}$ |

($\tau = 0.928$). The gap to ADAGE is 5.9 points. This is smaller than the gaps on the other datasets (42.5 to 44.6 points), and it is similar in size to the standard deviation of either method (8.0 and 5.7 points). With five seeds, this gap cannot be called reliable. Pubmed has only three classes, which gives it a higher random-guessing floor of about 33%. Its community structure may also let an attacker cover the graph with fewer queries. DP-ADAGE leaves some residual accuracy on other datasets as well. On Cora, 26.2% is still above the guessing level of about 14% for seven classes.

The $\tau$ column needs one caveat. PRADA and DAWN report $\tau = 0.000$ on every dataset, but this does not mean that attackers fail to diversify their queries. These two defenses use detection signals that differ from the diversity score, so the column does not apply to them. The same holds for NoDefense and StaticNoise, which track no diversity score. The accuracy and fidelity columns remain the right basis to compare all six defenses. For the two ADAGE-based defenses, $\tau$ is almost the same on each dataset (for example, 0.626 and 0.656 on Cora), because both measure the same query stream.

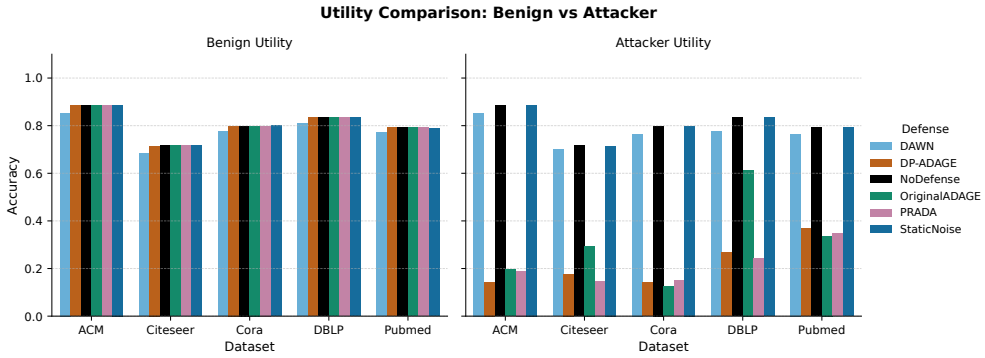



**Fig. 2** Benign accuracy (left) and attacker accuracy (right) across all five datasets for NoDefense, StaticNoise, ADAGE, PRADA, DAWN, and DP-ADAGE (mean over five seeds at query 200, error bars show one standard deviation). DP-ADAGE tracks NoDefense on benign accuracy and gives the lowest attacker accuracy on every dataset.

## 5.3 Benign vs. Attacker Utility

Figure 2 shows benign and attacker accuracy side by side for all five datasets and all six defenses. In the left panel, the DP-ADAGE, StaticNoise, ADAGE, and PRADA bars sit almost on top of the NoDefense bar on every dataset. The DAWN bar is visibly shorter, by about 3 points on every dataset. In the right panel, the DP-ADAGE bar is the shortest in every group. The PRADA and ADAGE bars have an intermediate height on most datasets. The DAWN bar tracks NoDefense closely because DAWN does little to slow an attacker. Together, the two panels show that only DP-ADAGE matches NoDefense on benign accuracy and also cuts attacker accuracy far below the undefended level.

## 5.4 Legitimate-User Experience Under Session Budgets

This subsection evaluates the protected API from the viewpoint of a legitimate user. Each benign session sent 200 sequential queries under a total privacy budget $\epsilon_{\text{total}} \in \{1, 5, 10, 200\}$. The earlier setting of $\epsilon_{\text{total}} = 200$ is kept only as a reference. The smaller budgets give a clearer view of the trade-offs among privacy, detection, and legitimate-user accuracy. The experiments use five datasets (ACM, CiteSeer, Cora, DBLP, and PubMed) and five random seeds.

Two availability metrics are reported. Transport availability is the fraction of queries that receive any response. Informative availability is the fraction of queries that receive a response before the session budget reaches its floor.

***Budget exhaustion.***

Table 3 summarizes the results for the session protocol with label outputs. Under $\epsilon_{\text{total}} = 200$, the budget reaches its floor after 21 queries on every dataset. Under $\epsilon_{\text{total}} \leq 10$, it reaches the floor after only 2 queries. Figure 3 shows the rolling accuracy of a legitimate user. For $\epsilon_{\text{total}} = 200$, accuracy stays close to 100% for about 21 queries.

**Table 3** Legitimate-user results under session budgets (label outputs; mean over five datasets and five seeds). The certificate $\epsilon^*$ follows Theorem 4.4. A certificate is meaningful when $\epsilon^* \leq 8.03$.

| $\epsilon_{\text{total}}$ | $\epsilon^*$ | Meaningful | Floor (queries) | Inform. avail. | Acc. @50 | Acc. @100 | Utility ret. |
|---|---|---|---|---|---|---|---|
| 1 | 2.97 | Yes | 2 | 0.005 | 0.203 | 0.200 | −0.001 |
| 5 | 6.97 | Yes | 2 | 0.005 | 0.208 | 0.214 | 0.014 |
| 10 | 11.97 | No | 2 | 0.005 | 0.208 | 0.214 | 0.014 |
| 200 | 201.97 | No | 21 | 0.100 | 0.529 | 0.372 | 0.126 |

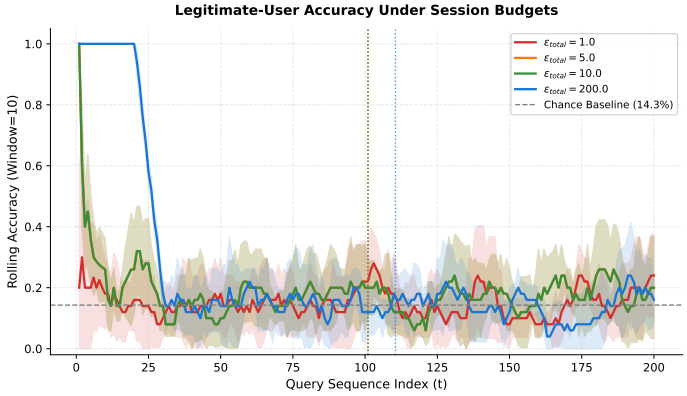



**Fig. 3** Rolling accuracy (window of 10 queries) of a legitimate user during a 200-query session. Shaded bands show variation across runs. The dashed line marks the chance level.

It then falls quickly and reaches the chance level of 14.3% at roughly query 30. For $\epsilon_{\text{total}} \leq 10$, accuracy is near chance from the start of the session.

***API availability.***

Transport availability is 1.0 in every setting, so the API always responds. However, informative availability is only 0.5% for $\epsilon_{\text{total}} \leq 10$ and 10% for $\epsilon_{\text{total}} = 200$. After the floor is reached, the service stays reachable but returns outputs with almost no useful signal. A client that monitors only response codes would therefore not detect the loss of utility.

***Privacy and utility.***

Only the two smallest budgets give a meaningful certificate ($\epsilon^* = 2.97$ and 6.97). In these cases, utility retention for session users is close to zero (Figure 4). The best retention in the session protocol is 0.126, at $\epsilon_{\text{total}} = 200$, where the certificate is not meaningful. Hence, no tested budget provides both a meaningful guarantee and a usable service for a session user. In contrast, the fresh protocol keeps 0.992 utility retention for label outputs at every budget. In that protocol, the budget does not accumulate across queries, so the result does not depend on $\epsilon_{\text{total}}$.

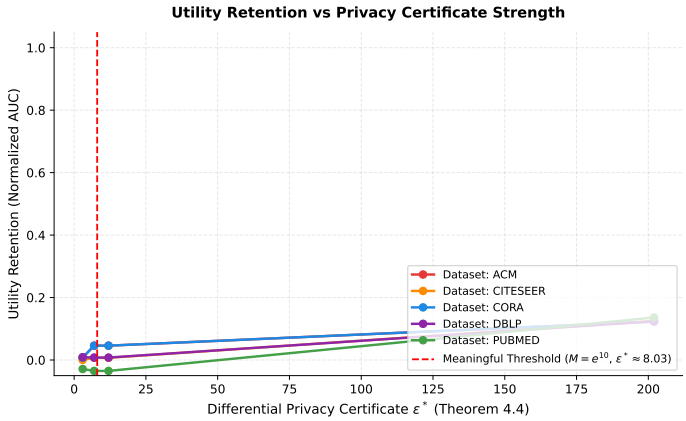



**Fig. 4** Utility retention versus the certificate $\epsilon^*$ for each dataset. The dashed line marks the threshold for a meaningful certificate.

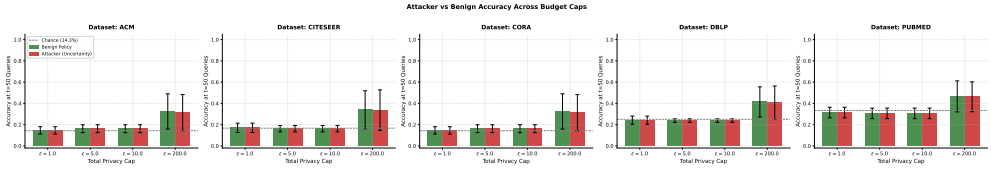



**Fig. 5** Accuracy after 50 queries for the benign policy and the uncertainty-based attacker under each budget cap. The dashed line marks the chance level.

***Legitimate users versus attackers.***

Figure 5 compares the accuracy of benign users and uncertainty-based attackers after 50 queries. On every dataset and at every budget, the two groups reach almost the same accuracy, and their error bars overlap. A session budget alone therefore does not separate attackers from legitimate users. It lowers the accuracy of both groups in the same way. Any separation must come from the detection mechanism and not from the budget cap.

***Implications.***

The results show a sharp loss of utility once the budget reaches its floor, and small budgets reach the floor almost immediately. A gradual noise increase or a periodic budget refresh for authenticated users may reduce this effect. The current experiments do not test these options, so they remain future work.

## 5.5 Surrogate Accuracy over the Query Budget

Figure 6 shows surrogate accuracy, averaged over the five datasets, as a function of the number of queries. The shaded bands show one standard deviation across seeds. The right axis shows the cumulative RDP budget $\varepsilon_{\text{cum}}$ tracked by the RDPAccountant,

together with the hard cap $\varepsilon_{\text{total}} = 200.0$. Under NoDefense, StaticNoise, and DAWN, mean surrogate accuracy is 73–76% at query 50 ($75.1 \pm 1.6$, $76.0 \pm 0.5$, and $73.2 \pm 1.8$) and ends at 78–80% at query 200. An attacker therefore needs only a few dozen queries to build a useful surrogate under these baselines. ADAGE and PRADA follow the same curve at first ($73.4 \pm 0.9\%$ and $75.6 \pm 1.3\%$ at query 50). They then decline, and end at $70.8 \pm 1.4\%$ and $67.5 \pm 2.3\%$ at query 200.

DP-ADAGE behaves differently. Its mean surrogate accuracy is $45.6 \pm 1.9\%$ at 10 queries (range 33.9–57.1% across datasets). It rises to a peak of $55.5 \pm 2.7\%$ at query 20, stays at the same level at query 30, and then falls as the accountant uses up the budget. At query 200, the mean is $34.5 \pm 2.6\%$ (range 26.2–49.0% across datasets). From query 30 on, DP-ADAGE has the lowest surrogate accuracy of the six defenses on all five datasets. At the start the order can be different. For example, on Cora DP-ADAGE is the highest of the six at query 10 (49.6%, against 36.8–46.5% for the others), while on Pubmed it is already the lowest (57.1%).

The budget is is exhausted early. With $\varepsilon_{\text{max}} = 10.0$ per query, the Citeseer and DBLP runs reach the cap of 200.0 by query 20 in every seed, and the Cora and ACM runs by query 30 in every seed. After that point the defense returns the query-independent output described in Section 3.3, so the curve cannot recover. On Pubmed, only one seed reaches the cap (at query 30). The mean spend on this dataset is 160.2 at query 30 and 162.9 at query 200.

Figure 7 gives the same curves for each dataset and adds PRADA and DAWN. NoDefense, StaticNoise, and DAWN stay together near the top of every panel. ADAGE and PRADA differ across datasets. On Cora and ACM, ADAGE peaks at query 90 to 100 (77.7% and 83.6%) and then declines to 70.7% and 75.8%. On Pubmed, ADAGE peaks much earlier, at query 40 (74.4%), and ends at 54.9%. On Citeseer and DBLP, ADAGE stays close to the undefended curve for the whole range, which explains why it barely helps in Table 2. PRADA follows the undefended curve until about query 100 on every dataset. It then drops to 65.3% (Cora), 59.1% (Citeseer), 71.4% (Pubmed), 70.7% (ACM), and 71.0% (DBLP) by query 200. On Pubmed, DP-ADAGE ends 5.9 points below ADAGE, the smallest margin among the datasets.

## 5.6 Query Budget Dynamics on Cora

Figure 8 focuses on Cora and shows all six defenses over the first 100 queries. The DP-ADAGE budget curve and the hard cap $\varepsilon_{\text{total}} = 200.0$ are overlaid. This case study comes from a separate set of five runs, so its curves differ from the Cora panel in Figure 7. For example, NoDefense reaches $68.8 \pm 5.9\%$ at query 100 here, against 78.9% in the longer run.

At 10 queries, the six defenses are indistinguishable. Their mean surrogate accuracy lies between 29.0% (StaticNoise) and 36.4% (NoDefense), and the standard deviation of each is between 10 and 16 points. DP-ADAGE is at $34.8 \pm 10.4\%$. The order changes within the first 50 queries. DP-ADAGE peaks at $44.2 \pm 7.6\%$ at query 30, and then falls to $40.9 \pm 5.5\%$ at query 50 and $34.3 \pm 2.5\%$ at query 100. The other five defenses reach between 52.4% (StaticNoise) and 63.9% (PRADA) at query 50, and between 61.9% and 68.8% at query 100. The gap at query 100 is therefore at least 27 points, far larger than any of the standard deviations.

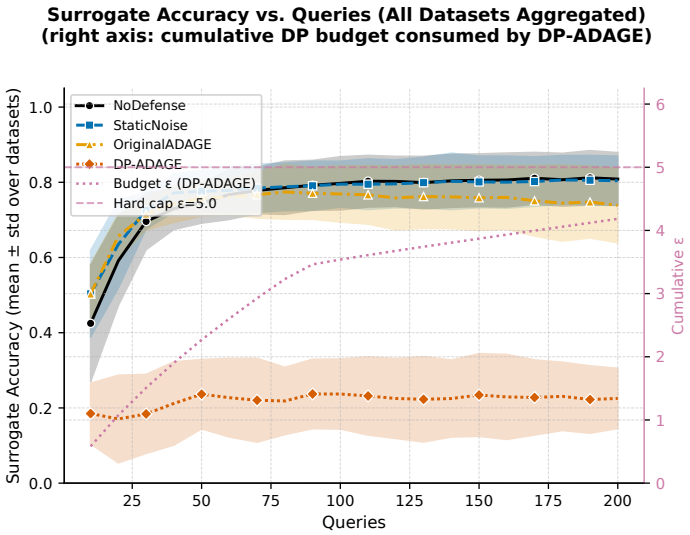



**Fig. 6** Surrogate accuracy versus number of queries, averaged over all five datasets (left axis, mean over five seeds with a band of one standard deviation). The right axis shows the cumulative RDP budget consumed by DP-ADAGE against the hard cap $\varepsilon_{\text{total}} = 200.0$. NoDefense, StaticNoise, and DAWN converge to a high surrogate accuracy within a few dozen queries, while DP-ADAGE keeps surrogate accuracy far lower for the full query budget.

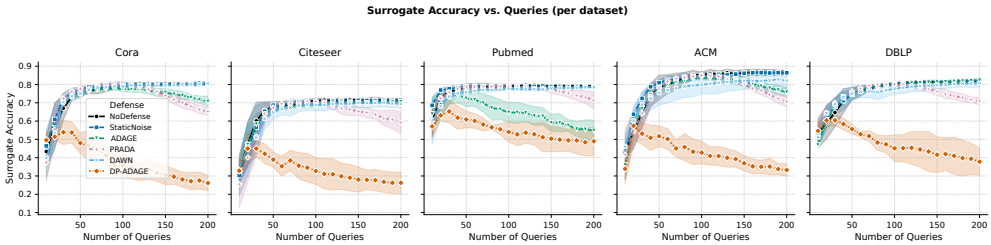



**Fig. 7** Surrogate accuracy versus number of queries for each of the five datasets, across NoDefense, StaticNoise, ADAGE, PRADA, DAWN, and DP-ADAGE (mean over five seeds, bands show one standard deviation). From query 30 on, DP-ADAGE has the lowest curve in every panel. The margin to ADAGE is smallest on Pubmed.

The DP-ADAGE budget line is at $99.6 \pm 0.1$ after 10 queries, at $197.5 \pm 0.3$ after 20, and reaches the full 200.0 by query 30. The accountant spends nearly all of its budget in the first 30 queries. This is what keeps the DP-ADAGE curve low afterward, even though it is not lower than the other defenses at the start.

## 5.7 Effect of the RDP Budget Cap

The main results use a loose cap of 200.0. Figure 9 repeats the aggregate query-budget experiment with two tighter caps, 1.0 and 10.0. Under the loose cap, mean surrogate accuracy starts at $41.4 \pm 3.7\%$ at query 10, peaks at $52.1 \pm 7.3\%$ at query 30, and then

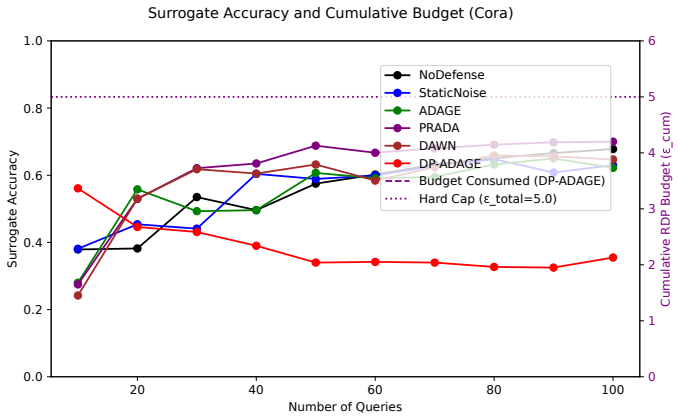



**Fig. 8** Surrogate accuracy and cumulative RDP budget on Cora over the first 100 queries for NoDefense, StaticNoise, ADAGE, PRADA, DAWN, and DP-ADAGE (mean over five seeds, bands show one standard deviation), with the DP-ADAGE budget curve and the $\varepsilon_{\text{total}} = 200.0$ hard cap overlaid. At query 10 all six defenses lie between 29% and 36%. DP-ADAGE reaches 34.3% at query 100, once its budget is used up, while the five baselines reach 62–69%.

declines to 32.6±1.9% by query 200 as the Noise Floor takes over. The two tighter caps behave very differently. With a cap of 10.0, mean surrogate accuracy stays between 19.4% and 22.7% for the whole range. With a cap of 1.0, it stays between 20.9% and 23.6%. Neither curve shows an early peak. The two tighter curves overlap, and the gap between them ($22.0 \pm 2.4\%$ against $20.5 \pm 1.0\%$ at query 200) is small compared with the gap to the loose cap.

The reason is the size of the per-query budget. Since $\varepsilon_{\max} = 10.0$, a cap of 1.0 or 10.0 is already used up by the first logged point at query 10. Every later response is then the uniform random label from the hard floor. On Cora, the surrogate accuracy under a cap of 1.0 is $13.1 \pm 1.8\%$, which is close to the 14% expected from random guessing over seven classes. A tighter cap removes the early window in which an attacker can reach a higher accuracy before the Noise Floor starts.

Table 4 shows the effect on benign accuracy at query 200. It does not change much across caps. The spread is at most 0.7 points on any dataset, and the largest is on ACM (87.0% at a cap of 1.0, 87.3% at 10.0, and 87.7% at 200.0). This spread is similar in size to the standard deviation of the individual cells (0.2 to 2.2 points). The dead-zone allocator is meant to explain this result. A benign user whose query diversity stays below $\tau_{\text{dead}}$ receives the maximum per-query budget, so the cap should not change the noise this user sees until the cap is used up.

This result requires a caveat. The benign column does not follow a benign user through a long session. A heavy benign user would also reach the Noise Floor once the cap is spent, as Section 3.3 points out. With a cap of 1.0 or 10.0 and a per-query budget of 10.0, this would happen after very few queries. The table therefore shows benign accuracy under the benign noise setting, not the cost of a tight cap to a high-volume legitimate user.

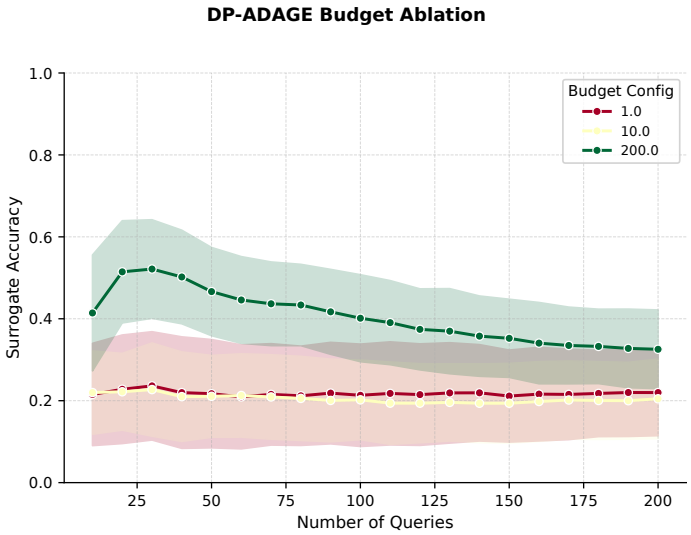



**Fig. 9** Surrogate accuracy versus number of queries for DP-ADAGE under three total RDP budget caps (1.0, 10.0, and 200.0), averaged over all five datasets (mean over five seeds, bands show one standard deviation). Both tighter caps remove the early peak in surrogate accuracy that appears under the loose cap.

For example, the two tight caps swap order across datasets: Citeseer is lower at a cap of 10.0 than at 1.0 ($14.7 \pm 3.0\%$ against $20.6 \pm 5.3\%$), although a tighter cap is not expected to raise attacker accuracy, while ACM shows the opposite order (13.1% at 1.0 and 14.3% at 10.0).

## 5.8 Utility–Security Trade-off

Figure 10 places the six defenses on a utility–security plane. The horizontal axis is benign accuracy, the vertical axis is one minus surrogate accuracy, and the bubble size shows query-time efficiency. Both axes use the mean over the five datasets at query 200. DP-ADAGE sits highest on the security axis, at $0.66 \pm 0.03$. Its mean benign accuracy is $80.5 \pm 0.4\%$, the same as NoDefense, ADAGE, PRADA, and StaticNoise (80.5–80.6%). It therefore lies closest to the empirical Pareto frontier. NoDefense, StaticNoise, and DAWN cluster at the low-security end, at 0.20–0.22. ADAGE ($0.29 \pm 0.01$) and PRADA ($0.33 \pm 0.02$) fall between the two extremes and do not reach the security level of DP-ADAGE. DAWN is the clear outlier on utility, with a mean benign accuracy of $77.7 \pm 0.5\%$. It sits to the left of every other defense without moving up the security axis, which confirms the weak trade-off seen in Figure 2.

## 5.9 Statistical Significance

Table 5 tests whether the lower surrogate accuracy of DP-ADAGE is a real effect. Each row is a paired $t$-test of DP-ADAGE against one baseline on the surrogate accuracy at query 200. The test uses the 25 pairs formed by five seeds and five datasets. The

**Table 4** Benign accuracy (%) and attacker surrogate accuracy (%) for DP-ADAGE under three RDP budget caps at query 200 (mean ± standard deviation over five seeds). Benign accuracy changes little across caps. Attacker accuracy is lowest under the two tighter caps, but those two are close to each other.

| | | Budget Cap | | |
|---|---|---|---|---|
| Dataset | Metric | 1.0 | 10.0 | 200.0 |
| Cora | Benign Acc. (%) | $80.6 \pm 0.6$ | $81.2 \pm 0.4$ | $80.9 \pm 0.6$ |
| | Att. Surr. Acc. (%) | $13.1 \pm 1.8$ | $12.2 \pm 1.7$ | $24.8 \pm 5.8$ |
| Citeseer | Benign Acc. (%) | $71.3 \pm 0.2$ | $71.3 \pm 0.4$ | $70.7 \pm 0.5$ |
| | Att. Surr. Acc. (%) | $20.6 \pm 5.3$ | $14.7 \pm 3.0$ | $26.7 \pm 1.6$ |
| Pubmed | Benign Acc. (%) | $79.4 \pm 0.3$ | $79.3 \pm 0.3$ | $79.1 \pm 0.3$ |
| | Att. Surr. Acc. (%) | $39.2 \pm 3.3$ | $35.4 \pm 4.9$ | $44.9 \pm 13.2$ |
| ACM | Benign Acc. (%) | $87.0 \pm 1.8$ | $87.3 \pm 2.2$ | $87.7 \pm 2.2$ |
| | Att. Surr. Acc. (%) | $13.1 \pm 4.2$ | $14.3 \pm 5.9$ | $30.9 \pm 3.7$ |
| DBLP | Benign Acc. (%) | $84.5 \pm 0.9$ | $84.6 \pm 0.7$ | $84.6 \pm 0.8$ |
| | Att. Surr. Acc. (%) | $23.9 \pm 5.5$ | $25.8 \pm 2.4$ | $35.4 \pm 2.8$ |

**Table 5** Paired $t$-tests of DP-ADAGE against each baseline on surrogate accuracy at query 200. The difference is the baseline minus DP-ADAGE, in percentage points. The first $p$-value uses the 25 seed–dataset pairs. The second uses the five seed means (4 degrees of freedom).

| Baseline | Mean diff. (pp) | $p$ (25 pairs) | $p$ (5 seed means) |
|---|---|---|---|
| NoDefense | 45.4 | $2.0 \times 10^{-17}$ | $2.7 \times 10^{-6}$ |
| StaticNoise | 45.3 | $3.6 \times 10^{-17}$ | $3.2 \times 10^{-6}$ |
| ADAGE | 36.3 | $1.1 \times 10^{-10}$ | $3.3 \times 10^{-6}$ |
| PRADA | 33.0 | $1.0 \times 10^{-15}$ | $1.9 \times 10^{-5}$ |
| DAWN | 43.8 | $3.3 \times 10^{-17}$ | $5.1 \times 10^{-6}$ |

mean difference ranges from 33.0 points (PRADA) to 45.4 points (NoDefense). All five $p$-values are below $10^{-9}$.

The 25 pairs are not fully independent, because five of them share each dataset. A stricter test uses the five seed means, each averaged over the datasets, with four degrees of freedom. This test gives $p$-values between $2.7 \times 10^{-6}$ and $1.9 \times 10^{-5}$. All five remain below 0.05 after a Bonferroni correction for five comparisons (threshold 0.01). The test supports the claim that DP-ADAGE lowers surrogate accuracy at query 200 in this setting. It does not say anything about other budgets, other attackers, or other datasets.

## 5.10 Budget-Aware Adaptive Attacker

Table 6 compares two attack strategies against each defense. The first is the standard uncertainty sampling used in the rest of this section. The second is a budget-aware

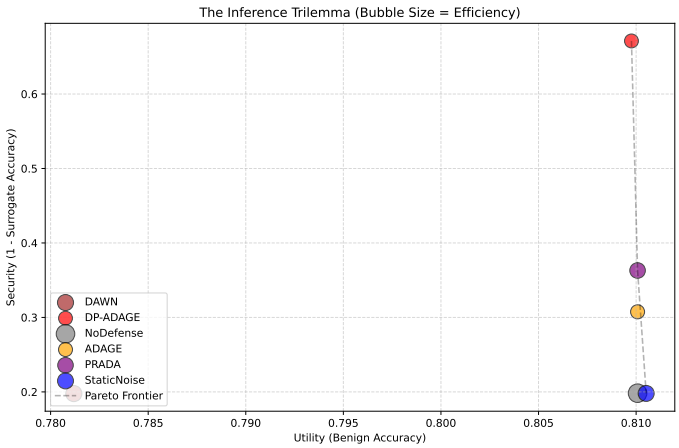



**Fig. 10** Utility (benign accuracy) versus security (1− surrogate accuracy) for NoDefense, StaticNoise, ADAGE, PRADA, DAWN, and DP-ADAGE, averaged over the five datasets at query 200 (mean over five seeds). Bubble size shows efficiency, and the dashed line shows the empirical Pareto frontier. DP-ADAGE reaches the highest security at nearly the same utility as NoDefense, StaticNoise, ADAGE, and PRADA, while DAWN gives up utility without a comparable security gain.

strategy, which is meant to use the queries more efficiently against a defense that has a hard cap. The runs extend to 500 queries, and each cell is the mean ± standard deviation over five seeds.

The budget-aware attacker is no more successful than the standard one against DP-ADAGE. At query 200 it reaches $26.2 \pm 4.8\%$ against $28.7 \pm 3.0\%$ for uncertainty sampling, and at query 500 the two are almost equal ($21.2 \pm 2.4\%$ and $21.0 \pm 2.1\%$). The DP-ADAGE budget is at the cap of 200.0 from query 30 on under both strategies. Both attackers therefore receive query-independent output for most of the run, and the accuracy keeps falling as more such output enters the training data. This is a likely reason for the decline, but the result files do not test it. The budget-aware attacker is also not stronger against NoDefense ($69.4 \pm 4.7\%$ against $73.3 \pm 3.6\%$ at query 500), so this attack does not seem to add much in this setup.

One result needs a careful reading. PRADA is no better than DP-ADAGE at query 200 against the budget-aware attacker ($31.3 \pm 6.6\%$ against $26.2 \pm 4.8\%$), and it is lower against the uncertainty attacker ($21.2 \pm 3.1\%$ against $28.7 \pm 3.0\%$). At query 500 it is lower under uncertainty sampling ($15.8 \pm 2.2\%$ against $21.0 \pm 2.1\%$) and almost equal under the budget-aware strategy ($20.7 \pm 2.6\%$ against $21.2 \pm 2.4\%$). PRADA starts to degrade the attack only after about 100 queries. Its surrogate accuracy under uncertainty sampling falls from 71.1% at query 100 to 39.8% at query 120. DP-ADAGE therefore has a clear advantage in the first 100 queries, and its advantage over PRADA does not hold for longer sessions in this experiment. ADAGE also improves with more queries (to $36.5 \pm 6.3\%$ at query 500 under uncertainty sampling), but it stays well above DP-ADAGE.

**Table 6** Surrogate accuracy (%) under two attack strategies at 200 and 500 queries (mean ± standard deviation over five seeds). Bold marks DP-ADAGE.

| | Uncertainty | | Budget-aware | |
|---|---|---|---|---|
| Defense | $Q = 200$ | $Q = 500$ | $Q = 200$ | $Q = 500$ |
| NoDefense | $72.7 \pm 3.6$ | $73.3 \pm 3.6$ | $68.3 \pm 6.1$ | $69.4 \pm 4.7$ |
| StaticNoise | $66.1 \pm 4.7$ | $66.5 \pm 3.1$ | $60.7 \pm 6.3$ | $64.4 \pm 3.8$ |
| ADAGE | $65.0 \pm 8.1$ | $36.5 \pm 6.3$ | $66.2 \pm 8.3$ | $43.8 \pm 10.1$ |
| PRADA | $21.2 \pm 3.1$ | $15.8 \pm 2.2$ | $31.3 \pm 6.6$ | $20.7 \pm 2.6$ |
| DAWN | $73.2 \pm 4.3$ | $72.9 \pm 3.9$ | $65.3 \pm 5.9$ | $68.3 \pm 5.2$ |
| **DP-ADAGE** | $\mathbf{28.7 \pm 3.0}$ | $\mathbf{21.0 \pm 2.1}$ | $\mathbf{26.2 \pm 4.8}$ | $\mathbf{21.2 \pm 2.4}$ |

**Table 7** Flickr results from a single run at 100 queries.

| Metric | NoDefense | DP-ADAGE |
|---|---|---|
| Surrogate acc. (%) | 42.3 | 42.3 |
| Budget at query 10 / 50 / 100 | N/A | 89.7 / 200.0 / 200.0 |
| Fit time (s) | $\approx 0$ | 25.1 |
| Avg. latency (ms) | 262.6 | 268.3 |

## 5.11 Scalability on Flickr

Table 7 reports a single run on the Flickr graph with up to 100 queries. This run measures cost on a larger graph. It cannot show protection, for the reason given below.

Under DP-ADAGE, the RDP budget grows by about 90 units in the first 10 queries and by about 34 units for every 10 queries after that, and it reaches the cap of 200.0 at query 50. The one-time fit of the defense takes 25.1 seconds, while NoDefense needs almost no setup time. The average latency per query is 268.3 ms for DP-ADAGE and 262.6 ms for NoDefense, an increase of 5.7 ms (2.2%).

The surrogate accuracy is the weak point of this run. It is 42.3% under NoDefense at every logged query, and it is 42.3% under DP-ADAGE (42.33% from query 90 on). A surrogate that does not change with the number of queries suggests that it did not learn from the responses, so this run cannot separate the two defenses. The cause was not investigated.

## 5.12 Query Latency, Memory, and Throughput

Table 8 lists the average per-query latency measured in the runs. Each cell is the mean ± standard deviation over five seeds. DP-ADAGE is slower than an unprotected API on every dataset. The ratio ranges from 1.33 on ACM to 2.07 on DBLP. ADAGE shows a similar overhead (1.32 to 2.05). Part of this cost may come from the community lookup in Equation 5, which compares each embedding with every centroid. StaticNoise, PRADA, and DAWN are also slower than NoDefense (for example, 329, 335, and 338 ms against 246 ms on Cora), so some of the overhead may come from the shared defense wrapper and not from DP-ADAGE alone. The ratio is lowest on ACM and highest on DBLP, which shows that it depends on the graph.

A separate profiling run measured the memory and throughput of DP-ADAGE. The per-user state takes 371.3 bytes, so 10,000 users need about 3.7 MB. The defense

**Table 8** Average per-query latency in milliseconds (mean ± standard deviation over five seeds). The ratio is DP-ADAGE latency divided by NoDefense latency, computed per seed.

| Dataset | NoDefense | ADAGE | DP-ADAGE | Ratio |
|---|---|---|---|---|
| Cora | $246 \pm 10$ | $372 \pm 11$ | $377 \pm 11$ | $1.54 \pm 0.08$ |
| Citeseer | $259 \pm 11$ | $424 \pm 19$ | $431 \pm 17$ | $1.67 \pm 0.09$ |
| Pubmed | $548 \pm 6$ | $982 \pm 14$ | $954 \pm 17$ | $1.74 \pm 0.04$ |
| ACM | $244 \pm 13$ | $322 \pm 17$ | $323 \pm 18$ | $1.33 \pm 0.13$ |
| DBLP | $630 \pm 2$ | $1288 \pm 10$ | $1302 \pm 10$ | $2.07 \pm 0.01$ |

serves 110.1 queries per second. Both numbers come from one measurement without repeated runs, so they give the order of magnitude and not a precise value.

### 5.13 Scope of the Evidence

The results above come from five seeds per configuration. They show large and consistent gaps between DP-ADAGE and the baselines: for example, the mean surrogate accuracy at query 200 is $34.5 \pm 2.6\%$ for DP-ADAGE, against $79.9 \pm 0.5\%$ for NoDefense. Gaps of a few points should not be read as real differences, because five seeds give only a rough estimate of the spread. Section 5.9 gives paired tests for the main claim.

Three limitations remain. First, the Flickr run uses one seed, and its surrogate accuracy does not respond to the number of queries, so it gives no evidence about protection on a large graph. Second, the adaptive-attacker test covers one budget-aware strategy, and PRADA matches or outperforms DP-ADAGE in long sessions there. Third, the latency and profiling numbers have no hardware description. A comparison of the Gaussian mechanism with Randomized Response is not reported, because the current result files contain no Gaussian-on-logits runs.

### 5.14 Multidimensional Comparison

Figure 11 compares the six defenses on four axes: security, utility, fidelity, and efficiency. DP-ADAGE extends farthest along the security axis and stays close to NoDefense on utility, as the accuracy numbers in Table 2 suggest. On the efficiency axis, DP-ADAGE and ADAGE give up the most ground, with mean latencies of about 680 ms across the five datasets, against 385 ms for NoDefense and 518–526 ms for StaticNoise, PRADA, and DAWN. PRADA covers a moderate area overall. It is ahead of StaticNoise and DAWN on security, and also ahead of ADAGE (0.33 against 0.29), but behind DP-ADAGE. DAWN covers the smallest area on the utility axis, which reflects its benign-accuracy cost.

## 6 Discussion and Limitations

### 6.1 Threat Model Scope

DP-ADAGE is designed to defend against *black-box model extraction attacks* where the adversary queries a GNN inference API and trains a surrogate model on the responses.

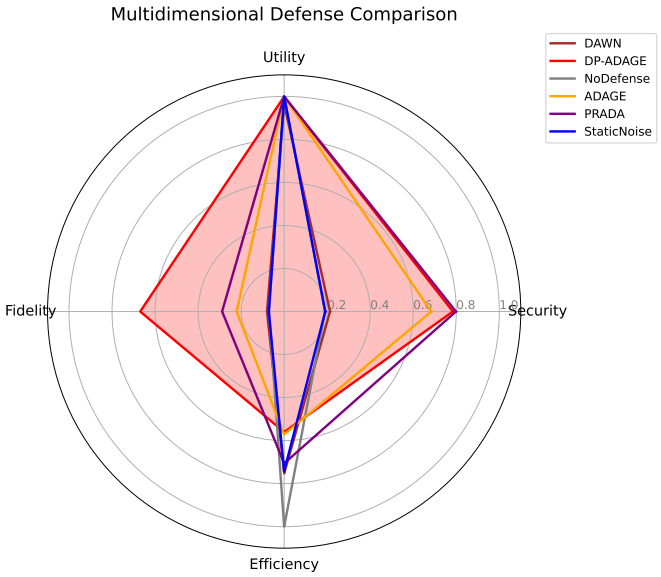



**Fig. 11** Multidimensional comparison of NoDefense, StaticNoise, ADAGE, PRADA, DAWN, and DP-ADAGE across security, utility, fidelity, and efficiency. DP-ADAGE covers the largest area on the security axis and stays competitive on utility.

The framework assumes the defender controls the inference endpoint, has access to the training graph (for community detection), and can track per-user query histories. It does not address white-box attacks (where model weights are directly accessible), training-data extraction, or side-channel attacks on the inference hardware. The experiments test two black-box strategies, uncertainty sampling and a budget-aware variant, on five citation-network benchmarks.

## 6.2 Limitations and Future Directions

***Scope of the Differential Privacy Guarantee.***

Differential privacy limits information leakage under defined assumptions. It does not automatically guarantee protection against model extraction. The guarantee bounds how much the distribution of the response transcript can change when two adjacent inputs differ, but it does not bound how well a surrogate model can approximate the victim function. This paper provides no theorem that links $\varepsilon^*$ to surrogate accuracy or fidelity, so protection against extraction is supported by experiments.

Noise makes knowledge distillation harder, but not impossible. A surrogate can still learn from noisy labels, and repeated queries allow an attacker to average Gaussian

noise or to take a majority vote over Randomized Response outputs. In the experiments, the drop in surrogate accuracy comes mainly from the hard budget cap. After the budget is exhausted, every response is independent of the query.

The certificate also depends on four assumptions. First, the output sensitivity is bounded by clipping with $C = B = 1.0$. Second, adjacency follows the definition in Section 4.2. Third, the budget is tracked per identity. Fourth, the approved per-query budget is computed from the diversity score $\tau$, which depends on unperturbed embeddings, and the privacy cost of this dependence is not accounted for. A natural next step is to derive a bound that links the transcript guarantee to surrogate error, and to compute $\tau$ under a private mechanism so that the fourth assumption can be removed.

***Budget Policy and Certificate Strength.***

The main utility result holds while a user has budget left. With $\varepsilon_{\max} = 10.0$ and $\varepsilon_{\text{total}} = 200.0$, a benign user who stays in the dead zone spends the whole budget after about 20 queries. The reported benign accuracy is measured with a fresh per-query budget, so it does not show how a high-volume legitimate user would fare in a long session. The cap also sets the strength of the certificate. The main setting gives $\varepsilon^* \approx 201.97$, which is valid but far above the threshold of about 8.0 from Proposition 1. The cap of 1.0 gives $\varepsilon^* \approx 2.97$ and lowers attacker accuracy to $22.0 \pm 2.4\%$, and the cap of 10.0 gives $20.5 \pm 1.0\%$. Both tighter caps are spent by the first logged query. Future work should evaluate a budget refill or reset policy, which would allow a single-digit $\varepsilon_{\text{total}}$ without cutting off legitimate users.

***Identity and Sybil Attacks.***

The per-user budget is tracked by identity (e.g., API key or IP address). An attacker who creates several identities receives a fresh budget for each one. The mean surrogate accuracy under DP-ADAGE reaches $55.5 \pm 2.7\%$ within the first 20 to 30 queries, and every new identity gives the attacker this early window again. This is a general property of per-user accounting and is not specific to DP-ADAGE. It can be reduced by rate limiting, CAPTCHAs, or payment-based access. A stronger direction is a global budget per queried node, which requires the defender to know the set of target nodes in advance.

***Adaptive Attackers.***

The experiments include one budget-aware attacker in addition to uncertainty sampling. Against DP-ADAGE, it reaches $26.2 \pm 4.8\%$ at 200 queries and $21.2 \pm 2.4\%$ at 500 queries, against $28.7 \pm 3.0\%$ and $21.0 \pm 2.1\%$ for uncertainty sampling. A fully adaptive attacker who knows $\tau_0$, $k_{\text{sharp}}$, and $\varepsilon_{\text{total}}$ could probe deeply within one community before switching, and may extract more information per budget unit. The RDP hard cap still bounds the total leakage in this case. The long-session results show a second point. At 500 queries, PRADA reaches $15.8 \pm 2.2\%$ under uncertainty sampling, below the $21.0 \pm 2.1\%$ of DP-ADAGE, so the advantage of DP-ADAGE is clearest in the first 100 queries. Future work should test stronger adaptive strategies and study

how DP-ADAGE and detection-based defenses such as PRADA can complement each other.

***Dynamic Graphs.***

The CommunityTracker runs Louvain once during offline preprocessing and caches the partition. The number of communities depends on the graph, from about 36 on Pubmed to over 600 on DBLP, so one new community moves $\tau$ by about 0.2% on DBLP and by 2–3% on Pubmed. The partition suits static graphs such as citation networks. In a dynamic graph, the centroids $\boldsymbol{\omega}_k$ drift and the diversity score may misrepresent query coverage. Incremental community detection, such as Louvain with incremental modularity updates, is a direction for extending DP-ADAGE to streaming graphs.

***Allocation and Sensitivity Calibration.***

The allocator parameters $k_{\text{sharp}} = 20$, $\tau_0 = 0.30$, and $\tau_{\text{dead}} = 0.05$ were chosen by hand, and the clipping norms $C = B = 1.0$ are a conservative bound that may over-noise when the true sensitivity is lower. The paper reports no sensitivity analysis for these values. Future work can study whether an optimal allocation function exists, either in closed form or learned from data [22], and can use data-dependent sensitivity estimates to reduce noise.

***Evaluation Scope.***

The main results use five seeds on citation networks, so differences of a few points should not be treated as reliable. The Flickr run uses a single seed and its surrogate accuracy does not respond to the number of queries, so it gives no evidence about protection on large graphs (Section 5.11). Repeating the large-graph experiment with a working attack and more seeds is the most direct extension of the evaluation.

# 7 Conclusion

This paper presented **DP-ADAGE**, a defense against model extraction attacks on graph neural network APIs that combines community-aware query-diversity tracking with a formal Rényi Differential Privacy (RDP) guarantee. Rather than relying on a fixed noise schedule, DP-ADAGE allocates the privacy budget with a dead-zone sigmoid function: a benign user whose queries stay within a small part of the graph keeps the full budget, while a user whose queries spread across many communities is pushed smoothly toward a strict budget as that spread grows. The perturbation step then matches the mechanism to the output type. Discrete label outputs are protected with Randomized Response, so the noise never overwhelms a signal that only has a handful of possible values, and continuous embedding outputs keep a calibrated Gaussian mechanism, which suits their high-dimensional structure. Both mechanisms compose under RDP into a single, auditable privacy certificate that covers an entire query session rather than one query at a time.

The experiments on five citation-network benchmarks, with five seeds each, support this design. With a total budget of 200, DP-ADAGE lowers the mean surrogate accuracy at 200 queries from $79.9 \pm 0.5\%$ to $34.5 \pm 2.6\%$, and a benign user's accuracy

stays within 0.1 points of the undefended API. Paired $t$-tests show that DP-ADAGE outperforms ADAGE, PRADA, DAWN, and static noise on this measure ($p < 10^{-4}$ in the strictest test). The budget is spent early: the cap is reached after 20 to 30 queries on four of the five datasets, and the first 100 queries are where DP-ADAGE gives the clearest advantage. The total RDP budget also works as a tunable parameter rather than a fixed setting. Caps of 1 and 10 lower the mean surrogate accuracy to $22.0 \pm 2.4\%$ and $20.5 \pm 1.0\%$, and they remove the early peak that appears under the loose cap. Only the cap of 1 gives a formal guarantee that is meaningful in practice ($\varepsilon^* \approx 2.97$). The cost of these tight caps to a high-volume legitimate user was not measured. Exposing this trade-off explicitly, instead of burying it inside an unprincipled heuristic, is what lets a deployer tune DP-ADAGE to their own risk tolerance.

Open challenges remain. In long sessions of 500 queries, PRADA matches or outperforms DP-ADAGE, and a Sybil attacker can restart the early window with a new identity. Extending DP-ADAGE to dynamic graphs would require streaming community detection instead of the current offline Louvain step. Hardening the defense against Sybil attacks would call for budget accounting that ties back to individual nodes rather than individual accounts. The evidence on large graphs is also missing, because the Flickr run did not produce a working attack. Finally, the sigmoid-based allocator, even with its dead zone, is a hand-tuned shape rather than a proven optimum, so deriving an allocation function with a formal utility guarantee is still an open problem. This paper's implementation and experimental pipeline are released to support reproducibility and to help others build on this direction at the intersection of graph privacy and machine learning security.

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